Small-energy finite-time singularities of Calabi flow

Abstract

We give a construction of smooth Calabi flows on complex projective seven-space, in the fixed Fubini–Study class, that develop a finite-time scalar-curvature singularity while their Calabi energies are arbitrarily small and their Sobolev constants obey one common bound. The initial Kähler potentials can be arbitrarily small in the uniform norm, and the singular time can be arbitrarily short. More generally the scalar curvature minus its average is arbitrarily small, uniformly in time, in any prescribed L^p norm with $2 \le p < 7$. The construction localizes a positive shrinking profile and solves the resulting fourth-order equation by an inner ancient inverse and an outer zero-data inverse. A finite rational certificate establishes the seven-dimensional profile.

1 Statement

Let ω_* denote the Fubini–Study form on $\mathbb{CP}^7$ whose affine potential is $\log(1+|Z|^2)$. Scalar curvature means $S = \mathrm{tr}_\omega \mathrm{Ric}(\omega)$, so $S(\omega_*) = 56$. Write $\mathrm{d}\nu_\omega = \omega^7 / \int_{\mathbb{CP}^7} \omega^7$. The Sobolev constant used here is the least C for which

$$\|f\|^2_{L^{7/3}(\nu_\omega)} \le C \left(\|\nabla f\|^2_{L^2(\nu_\omega)} + \|f\|^2_{L^2(\nu_\omega)} \right) \qquad (f \in C^\infty(\mathbb{CP}^7; \mathbb{R})). \tag{1.1}$$

The underlying Riemannian dimension is fourteen.

Theorem 1.1. *There is $C_S < \infty$ with the following property. For every $2 \le p < 7$ and $\varepsilon, \tau > 0$, there is a smooth real $U(7)$-invariant potential ϕ_0 such that $\omega_0 = \omega_* + i\partial\bar\partial\phi_0 > 0$ and the maximal smooth Calabi flow from ω_0 has existence interval $[0, T_*)$ satisfying*

$$\|\phi_0\|_{C^0} < \varepsilon, \qquad 0 < T_* < \tau, \tag{1.2}$$

$$\sup_{0 \le t < T_*} \|S(\omega(t)) - 56\|_{L^p(\nu_{\omega(t)})} < \varepsilon, \tag{1.3}$$

$$C_S(\omega(t)) \le C_S \quad (0 \le t < T_*). \tag{1.4}$$

There are a fixed point o and $a > 0$ such that

$$S(\omega(t))(o) \sim a(T_* - t)^{-1/2} \qquad (t \uparrow T_*). \tag{1.5}$$

The constant C_S is independent of p, ε, τ and t.

Taking $p = 2$ gives arbitrarily small Calabi energy throughout the smooth existence interval. The class and volume remain fixed, so this smallness does not result from scaling the entire metric. Small initial energy together with the common bound (1.1) therefore does not guarantee smooth long-time existence in this class.

The proof has three substantive parts: the positive profile in Section 3, the localized correction in Section 6, and the uniform geometric estimates in Section 7. The analytic inverse is established in Sections 4–5. Standard fixed-point, spectral, inverse-function and smooth parabolic existence theorems are used. All estimates whose constants must be uniform in the shrinking spatial and temporal parameters are recorded below. The finite computation in Section 3 is an explicit part of the proof; it is not a numerical simulation of the flow.

2 The geometric reduction and the finite-time obstruction

We use scalar curvature in the convention $S(\omega) = \mathrm{tr}_\omega \mathrm{Ric}(\omega)$. Let ω_* be the Fubini–Study form on $\mathbb{CP}^n$ with affine potential $\log(1+|Z|^2)$. The geometric identities below hold for every $n \geq 1$; the construction will use $n = 7$. We give the identities directly to fix the scalar convention and to verify smoothness at the point orbit as well as at the divisor. Set

$$q = |Z|^2, \qquad \ell = \log q, \qquad v_*(x) = x(1-x), \qquad w(x) = x^{n-1}. \tag{2.1}$$

For a radial potential $\Phi(\ell)$, its momentum coordinate and momentum profile are

$$x = \Phi'(\ell), \qquad v(x) = \Phi''(\ell).$$

The Legendre transform $u(x) = x\ell - \Phi(\ell)$ satisfies $u' = \ell$ and $u'' = 1/v$. The reference transform is

$$u_*(x) = x\log x + (1-x)\log(1-x), \qquad u_*'' = v_*^{-1}.$$

Primes on functions of x will denote differentiation in x.

Proposition 2.1 (Exact fixed-class reduction)**.** *Let I be an open time interval, and let $H \in C^\infty(I \times [0,1])$ satisfy*

$$b(t,x) := 1 + v_*(x)H(t,x) > 0. \tag{2.2}$$

Define

$$v = v_* + v_*^2 H = v_* b, \qquad K(H) = w^{-1}(wv_*^2 H)'', \qquad \mathcal{A}H = \left[w^{-1}(wv_*^2 H)''\right]''. \tag{2.3}$$

At each time these data, together with an affine choice in the Legendre potential, determine a smooth $U(n)$-invariant Kähler form in $[\omega_*]$. Its scalar curvature and average scalar curvature are

$$S(t,x) = n(n+1) - K(H)(t,x), \qquad \bar{S} = n(n+1). \tag{2.4}$$

If, in addition,

$$H_t + (1+v_*H)^2 \mathcal{A}H = 0, \tag{2.5}$$

then for any $t_0 \in I$ and any initial affine choice there is a smooth family of these forms satisfying the Calabi flow exactly. More precisely, its global potential relative to $\omega(t_0)$ satisfies

$$\phi(t_0) = 0, \qquad \omega(t) = \omega(t_0) + i\partial\bar\partial\phi(t), \qquad \phi_t = S(\omega(t)) - \bar{S}. \tag{2.6}$$

Proof. We first verify smoothness and positivity at both endpoints. Suppress time for this part of the argument. Since

$$\frac{1}{v} - \frac{1}{v_*} = -\frac{H}{1+v_*H}, \tag{2.7}$$

we can choose $\psi \in C^\infty[0,1]$ with second derivative equal to the right-hand side, and put $u = u_* + \psi$. The two integration constants are precisely its free affine part. Write $a = \psi'$. The function u' is strictly increasing from $-\infty$ to $+\infty$, so it has a smooth inverse on the open interval. In terms of $q = \exp\ell$, the inverse relation is

$$q = \frac{x}{1-x}\exp a(x). \tag{2.8}$$

At $x = 0$ its right-hand side vanishes and has derivative $\exp a(0) > 0$. The inverse therefore extends smoothly to $q = 0$, with

$$x(q) = \exp(-a(0))q + O(q^2).$$

Here and below a smooth remainder is meant: for example the last error is q^2 times a smooth function of q. Since $\mathrm{d}\Phi/\mathrm{d}q = x(q)/q$, integration shows that the Legendre dual extends smoothly as a function of q, with expansion

$$\Phi(\log q) = C_0 + \exp(-a(0))q + O(q^2). \tag{2.9}$$

Division by q here preserves smoothness because $x(0) = 0$, as is also clear from $x(q)/q = \int_0^1 x'(sq)\,\mathrm{d}s$. Its complex Hessian at $Z = 0$ is $\exp(-a(0))$ times the Euclidean Hermitian form, and is positive.

At the other endpoint set $\eta = q^{-1}$ and $b_1 = 1 - x$. The same inverse relation becomes

$$\eta = \frac{b_1}{1-b_1}\exp\big(-a(1-b_1)\big), \qquad b_1(\eta) = \exp a(1)\eta + O(\eta^2).$$

If $\Psi(\eta) = \Phi(\log q) - \log q$, then $\Psi'(\eta) = b_1(\eta)/\eta$. Consequently

$$\Psi(\eta) = C_1 + \exp a(1)\eta + O(\eta^2) \tag{2.10}$$

is smooth at $\eta = 0$. This expansion gives smoothness on the projective manifold, not merely along radial rays. Indeed, on the projective chart meeting the divisor at infinity given by

$$W_0 = 1/Z_j, \qquad W_i = Z_i/Z_j \quad (i \neq j), \qquad B = 1 + \sum_{i\neq j} |W_i|^2,$$

we have $q = B/|W_0|^2$. After removing the pluriharmonic term $-\log|W_0|^2$ from Φ, a local potential is

$$\log B + \Psi\left(\frac{|W_0|^2}{B}\right). \tag{2.11}$$

On $W_0 = 0$ its tangential complex Hessian is the Fubini–Study metric of the divisor, its normal coefficient is $\exp a(1)/B > 0$, and its mixed coefficients vanish. This proves positivity there; for $n = 1$ only the normal coefficient is present. On $0 < x < 1$, direct differentiation gives the tangential and radial eigenvalues x/q and v/q, respectively, so positivity holds everywhere.

The difference $\Phi(\log|Z|^2) - \log(1+|Z|^2)$ is a globally smooth function. At the origin this follows from (2.9); in the chart at infinity it equals $\Psi(\eta) - \log(1+\eta)$, which is smooth by (2.10). Thus its complex Hessian is a global exact form and the Kähler class is $[\omega_*]$. All inversions and integrations above depend smoothly on time on compact subintervals of I, by the positive endpoint derivatives and the inverse function theorem with parameters.

We next compute scalar curvature. The eigenvalues already found give

$$\det(g_{i\bar{j}}) = q^{-n}x^{n-1}v = q^{-n}wv.$$

For any radial function f, the complex Laplacian is

$$\Delta_\omega f = \frac{f_{\ell\ell}}{v} + \frac{n-1}{x}f_\ell = w^{-1}\partial_x(wf_\ell), \tag{2.12}$$

where $\partial_\ell = v\partial_x$ in the last expression. For $f = -\log\det g$, this yields

$$f_\ell = n - \frac{(wv)'}{w}, \qquad S = \frac{n(n-1)}{x} - w^{-1}(wv)''.$$

Substituting $v = v_* + v_*^2 H$ gives (2.4). The expression $K(H)$ is smooth at zero: $wv_*^2 H = x^{n+1}(1-x)^2 H$, whose second derivative is divisible by x^{n-1} as a smooth function. It is also smooth at one.

For completeness, the average can be read directly from this formula. The radial volume density is a constant multiple of $w(x)\,\mathrm{d}x$: the Euclidean polar factor $q^{n-1}\mathrm{d}q$, multiplied by the determinant, becomes $wv\,\mathrm{d}\ell = w\,\mathrm{d}x$. Moreover,

$$\int_0^1 K(H) w\,\mathrm{d}x = \left[(wv_*^2 H)'\right]_{x=0}^{x=1} = 0,$$

because $wv_*^2 H$ vanishes to order at least two at both endpoints. This proves the asserted average, including when $n = 1$.

Finally, fix $t_0 \in I$, choose $u(t_0)$ as above, and define at fixed momentum coordinate

$$u(t,x) = u(t_0,x) + \int_{t_0}^t K(H)(s,x)\,\mathrm{d}s. \tag{2.13}$$

Equation (2.5) implies

$$\partial_t(1/v) = -\frac{v_*^2 H_t}{v^2} = \mathcal{A}H = \partial_x^2 K(H) = \partial_t u''.$$

Their initial values agree, so $u'' = 1/v$ for all time. In particular the preceding metric construction applies throughout I. Differentiating the Legendre dual $\Phi(t,\ell) = x(t,\ell)\ell - u(t,x(t,\ell))$ at fixed ℓ cancels the terms containing x_t, and gives

$$\Phi_t = -u_t = -K(H) = S - \bar{S}. \tag{2.14}$$

The difference between $\Phi(t)$ and $\Phi(t_0)$ is globally smooth by the endpoint argument and therefore supplies ϕ in (2.6).

This also specifies the affine part of u during evolution. Replacing a fixed u by $u + \alpha x + \beta$ changes its dual to $\Phi(\ell - \alpha) - \beta$, corresponding to the projective dilation $Z \mapsto \exp(-\alpha/2)Z$ and a potential constant. An arbitrary time-dependent choice of these terms would affect (2.14); the prescription (2.13) fixes them and establishes that equation without an additional affine term. □

2.1 A regular auxiliary sphere and its flat limit

The endpoints of the momentum interval can be handled by a smooth equation on a closed auxiliary manifold. Set

$$m = 2n + 4, \qquad \Sigma = S^{m+5} \subset \mathbb{R}^m \times \mathbb{R}^6, \qquad x = |z|^2. \tag{2.15}$$

We identify functions of x with $O(m) \times O(6)$-invariant functions on Σ.

Lemma 2.2 (The compact and flat operators)**.** *This identification is a bijection between $C^\infty[0,1]$ and the smooth invariant functions on Σ. The pushforward of spherical volume under x is a positive constant multiple of $wv_*^2\,\mathrm{d}x$. With the nonpositive Laplacian convention, put $L = \frac{1}{4}\Delta_\Sigma$. On invariant functions,*

$$L = x(1-x)\partial_x^2 + (n+2-(n+5)x)\partial_x, \qquad \mathcal{A} = (-L+n+3)(-L+2(n+2)). \tag{2.16}$$

The right-hand expression defines a smooth elliptic operator of order four on all of Σ.

In the core coordinates $(z,\theta) \mapsto (z, \sqrt{1-|z|^2}\,\theta)$, with $\theta \in S^5$, its Laplacian on functions independent of θ is

$$\Delta_\Sigma = \Delta_z - \sum_{i,j=1}^m z_i z_j \partial_{z_i z_j} - (m+5)\sum_{i=1}^m z_i \partial_{z_i}. \tag{2.17}$$

The flat momentum operator obtained by replacing v_* by x in (2.3) satisfies

$$\widehat{\mathcal{A}}H := \left[x^{1-n}(x^{n+1}H)''\right]'' = \left(x\partial_x^2 + (n+2)\partial_x\right)^2 H = \frac{1}{16}\Delta_{\mathbb{R}^m}^2 H, \qquad x = |z|^2. \tag{2.18}$$

Proof. The only regularity issue in the identification occurs at the two singular orbits. Here is an elementary smoothness argument. If $f(r)$ is smooth and even near zero, then the operator

$$\mathcal{T}f(r) = \frac{f'(r)}{r} = \int_0^1 f''(sr)\,\mathrm{d}s$$

defines another smooth even function. Thus, for $g(x) = f(\sqrt{x})$, each derivative on $x > 0$ has the form $g^{(j)}(x) = 2^{-j}(\mathcal{T}^j f)(\sqrt{x})$, and extends continuously to $x = 0$. Induction gives smoothness up to zero of g and all its derivatives. The same argument holds with smooth parameters. Restricting a smooth invariant function on Σ to $(re_1, \sqrt{1-r^2}\,\theta_0)$ gives such an even f, and invariance makes its value depend only on $x = r^2$. Interchanging the two factors proves smoothness in $1 - x$ at the other endpoint. Conversely a smooth function of x is smooth in these coordinates at both orbits, as well as on their complement.

Write $|z| = \sin\vartheta$, with $0 \le \vartheta \le \pi/2$. The spherical metric and volume density are

$$\mathrm{d}\vartheta^2 + \sin^2\vartheta\, g_{S^{m-1}} + \cos^2\vartheta\, g_{S^5}, \qquad \sin^{m-1}\vartheta\cos^5\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\omega_{m-1}\,\mathrm{d}\omega_5.$$

Changing to $x = \sin^2\vartheta$ gives density proportional to

$$M(x)\,\mathrm{d}x := x^{m/2-1}(1-x)^2\,\mathrm{d}x = wv_*^2\,\mathrm{d}x.$$

The radial Laplacian computed from this metric gives the displayed formula for L, equivalently $L = M^{-1}\partial_x(v_*M\partial_x)$.

We verify the factorization by a polynomial argument that also checks the endpoint terms. Both L and $\mathcal{A}$ preserve polynomial degree and are symmetric for the inner product with density M. For $\mathcal{A}$, twice integrating by parts gives

$$\int_0^1 a(\mathcal{A}b)M\,\mathrm{d}x = \int_0^1 \frac{(Ma)''(Mb)''}{w}\,\mathrm{d}x. \tag{2.19}$$

There are no boundary terms: Ma and its first derivative vanish at each endpoint, and $w^{-1}(Mb)''$ is smooth there. For L symmetry follows from its divergence expression, since v_*M vanishes at both ends. Orthogonalizing the monomials therefore gives common polynomial eigenfunctions. On the polynomial of degree j, the leading coefficients give respectively the eigenvalues

$$-j(j+n+4), \qquad (j+1)(j+2)(j+n+2)(j+n+3).$$

Indeed a symmetric degree-preserving operator maps the degree-j orthogonal polynomial to a multiple of itself, because its inner product with every lower-degree polynomial vanishes. Finally,

$$\big(j(j+n+4)+n+3\big)\big(j(j+n+4)+2(n+2)\big) = (j+1)(j+n+3)(j+2)(j+n+2).$$

This proves equality on every polynomial, hence equality of the differential expressions: at an interior point polynomials realize arbitrary jets through order four. The sphere expression has leading term $\Delta_\Sigma^2/16$, proving the claimed ellipticity.

For (2.17), the inverse metric in the z coordinates is $(\delta_{ij} - z_iz_j)$ and the volume density is $(1-|z|^2)^2\,\mathrm{d}z\,\mathrm{d}\omega_5$. Applying the divergence formula therefore gives

$$(1-|z|^2)^{-2}\partial_{z_i}\big((1-|z|^2)^2(\delta_{ij}-z_iz_j)\partial_{z_j}\big) = (\delta_{ij}-z_iz_j)\partial_{z_iz_j} - (m+5)z_i\partial_{z_i},$$

with repeated indices summed. This computation only requires independence of θ. Finally both one-dimensional expressions for $\widehat{\mathcal{A}}$ in (2.18) expand to

$$x^2H^{(4)} + 2(n+3)xH^{(3)} + (n+2)(n+3)H'',$$

and the radial Euclidean identity $\frac{1}{4}\Delta_{\mathbb{R}^m} = x\partial_x^2 + \frac{m}{2}\partial_x$ proves the last equality. □

2.2 The self-similar core

From now on set $n = 7$, so $m = 18$ and $\Sigma = S^{23}$. In the flat model the equation for the coefficient is

$$F_t + (1 + xF)^2 \widehat{\mathcal{A}} F = 0. \tag{2.20}$$

Let $D = x\partial_x$. For the function $g = 1 + xF$, (2.18) gives

$$x\widehat{\mathcal{A}}(x^{-1}(g-1)) = x^{-2}P(D)(g-1), \qquad P(D) = (D-2)(D-1)(D+6)(D+7). \tag{2.21}$$

Here P denotes a polynomial, as distinct from the evolution operator denoted by P later. One can check (2.21) on $g - 1 = x^j$: both sides equal $(j-2)(j-1)(j+6)(j+7)x^{j-2}$; equality of the differential expressions follows as in Lemma 2.2. Consequently (2.20) is equivalent, for $x > 0$, to

$$g_t = -g^2 x^{-2} P(D)(g-1). \tag{2.22}$$

Define the cubic polynomial operator $Q(D) = (D-1)(D+6)(D+7)$. Suppose that G is smooth and positive on $[0, \infty)$, with $G(0) = 1$, and for some $c > 0$ satisfies

$$Q(y\partial_y)(G-1) = \frac{y^2}{2}\left(\frac{1}{G} - \frac{1}{c}\right). \tag{2.23}$$

Then set

$$\rho = \sqrt{-t}, \qquad y = x/\rho, \qquad h(y) = \frac{G(y)-1}{y}, \qquad F(t,x) = \rho^{-1}h(x/\rho), \qquad t < 0. \tag{2.24}$$

The value $h(0) = G'(0)$ makes h smooth at zero, since $h(y) = \int_0^1 G'(sy)\,\mathrm{d}s$. Thus F is smooth for $t < 0$ including at the flat origin. Applying $D - 2$, now with $D = y\partial_y$, to (2.23) yields

$$P(D)(G-1) = -\frac{y^2 DG}{2G^2}.$$

On the other hand $\rho_t = -1/(2\rho)$, so $\partial_t G(x/\rho) = DG/(2\rho^2)$. These identities prove (2.22), and hence (2.20), first for $x > 0$ and then at zero by smoothness. The existence of the required nonconstant positive G is the subject of Theorem 3.1.

The same expression $F(t,x)$ is smooth on the compact sphere for every $t < 0$. Its compact positivity multiplier obeys the useful identity

$$1 + v_* F = x + (1-x)G(x/\rho). \tag{2.25}$$

In particular, if G is bounded above and below by positive constants, this multiplier has the same property uniformly in $t < 0$ and $0 \le x \le 1$. This is a bound in the momentum coordinate x, whose relation to the fixed affine coordinate varies under the Legendre transform. It does not imply uniform equivalence to the Fubini–Study metric in the fixed complex coordinates.

2.3 Curvature at the concentrating point

The final geometric implication uses only one value of the correction. It is convenient to state it before constructing that correction.

Lemma 2.3 (Finite-time obstruction)**.** *Let $G \in C^\infty[0,\infty)$ satisfy $G(0) = 1$ and $G'(0) \neq 0$, and define F by (2.24). Suppose that for some $T > 0$ and $\delta > 0$ there is a smooth invariant solution H of (2.5) on $(-T, 0) \times \Sigma$ such that $1 + v_* H > 0$, $H(t, 0) = F(t, 0) + V(t, 0)$, and*

$$V(t, 0) = O(\rho^{-1+\delta}) \qquad (t \uparrow 0). \tag{2.26}$$

Then a smooth $U(7)$-invariant metric $\omega_0 \in [\omega_*]$ has maximal smooth Calabi-flow existence interval $[0, T_*)$, where $T_* = T/2 < \infty$. At the fixed affine origin p, the scalar curvature of this flow satisfies

$$S(\tau, p) = a(T_* - \tau)^{-1/2} + O\big(1 + (T_* - \tau)^{(-1+\delta)/2}\big), \qquad a = -56G'(0) \neq 0. \tag{2.27}$$

In particular $S(\tau, p) \sim a(T_* - \tau)^{-1/2}$, and $a > 0$ when $G'(0) < 0$.

Proof. By Lemma 2.2, H is smooth in x at both endpoints for each time. Proposition 2.1, with $t_0 = -T/2$, gives a smooth family of Kähler forms, and its slice $\omega_0 = \omega(-T/2)$ is a smooth positive form in the Fubini–Study class. Put $\tau = t + T/2$. The corresponding global potential satisfies exactly the normalization and evolution in (2.6) on $0 \leq \tau < T_* = T/2$.

At $x = 0$, the coefficient of x^{n+1} in $wv_*^2 H$ is $H(t, 0)$. Thus (2.4) gives the exact identity

$$S(t, p) = n(n + 1)\big(1 - H(t, 0)\big). \tag{2.28}$$

This evaluates scalar curvature at the same point p at every time: (2.8) sends $q = 0$ to $x = 0$ for every affine choice. Since

$$H(t, 0) = \rho^{-1} G'(0) + O(\rho^{-1+\delta}),$$

Equation (2.28) proves (2.27) and the asserted divergence.

We recall why uniqueness applies to comparison with an unrestricted smooth flow; no symmetry hypothesis is needed for this step. Local existence follows from the smooth fourth-order parabolic Cauchy theorem. The uniqueness mechanism is as follows. For two smooth solutions with the same initial potential, let χ be their difference and interpolate their potentials by φ_s, $0 \leq s \leq 1$. Their interpolated Kähler forms remain positive, and variation of scalar curvature gives

$$\chi_t = -\int_0^1 \Big(\Delta_{\varphi_s}^2 \chi + \mathrm{Ric}_{\varphi_s}^{i\bar{j}}\, \partial_i \partial_{\bar{j}} \chi\Big)\, \mathrm{d}s.$$

On a compact time interval where both solutions are smooth, the coefficients are smooth and uniformly elliptic. Relative to a fixed smooth volume form, integration twice by parts and interpolation give

$$\frac{\mathrm{d}}{\mathrm{d}t} \|\chi\|_{L^2}^2 \leq -c\|\chi\|_{H^2}^2 + C\|\chi\|_{L^2}^2 \leq C\|\chi\|_{L^2}^2.$$

For this estimate, the principal quadratic form is the integral of $|\Delta_{\varphi_s}\chi|^2$; differentiating the smooth volume density introduces terms bounded by $C\|\chi\|_{H^1}\|\chi\|_{H^2}$, and the Ricci term has the same or lower order. The elliptic estimate for each uniformly elliptic Laplacian and $\|\chi\|_{H^1} \leq \varepsilon\|\chi\|_{H^2} + C_\varepsilon\|\chi\|_{L^2}$ absorb those terms. Gronwall's inequality proves uniqueness.

It follows that any smooth flow from ω_0 agrees with the one constructed above on their common interval. Hence its maximal interval contains $[0, T_*)$. If it extended beyond T_*, smooth positivity on the compact manifold across that time would bound its scalar curvature, contradicting (2.27). Its maximal smooth existence time is therefore exactly T_*. □

3 A positive seven-dimensional shrinking profile

Theorem 3.1. *There are $0 < c < 1$ and $G \in C^\infty[0,\infty)$ such that*

$$(D-1)(D+6)(D+7)(G-1) = \frac{y^2}{2}(G^{-1} - c^{-1}), \qquad D = y\partial_y, \tag{3.1}$$

$$G(0) = 1, \quad G'(0) < 0, \quad 0 < \inf G \le \sup G < \infty. \tag{3.2}$$

For every $J, \ell \ge 0$, there are coefficients g_j with

$$\partial_y^\ell \left(G(y) - c - \sum_{j=1}^{J} g_j y^{-2j} \right) = O(y^{-2(J+1)-\ell}). \tag{3.3}$$

In particular $F(t,x) = \rho^{-1}(G(x/\rho) - 1)/(x/\rho)$ solves $F_t + (1+xF)^2 \widehat{\mathcal{A}} F = 0$, including at $x = 0$.

3.1 The equation and the analytic family at zero

Put

$$c = (1+d)^{-1}, \qquad s = (27/16)^{1/3}(y/c)^{2/3}, \qquad G(y) = c(1 + p(s)). \tag{3.4}$$

Direct substitution in (3.1) yields

$$p''' + \frac{21}{s}p'' + \frac{B}{s^2}p' + \frac{C}{s^3}(p-d) + \frac{p}{1+p} = 0, \qquad B = 337/4, \quad C = -567/4. \tag{3.5}$$

Set $r = s^{3/2}$. An analytic solution germ has the form

$$p = d + kr + \sum_{i \ge 2} a_i r^i, \qquad a_i = \frac{[r^{i-2}]((1+p)^{-1} - 1)}{(27/8)(i-1)(i+6)(i+7)}. \tag{3.6}$$

The exact rational center (k_0, d_0) and matrix M used below are specified by terminating decimals in Table 1. Their method of discovery is immaterial. Let

$$R = 10^{-12}, \qquad z \in [-R, R]^2, \qquad (k,d)^t = (k_0, d_0)^t + Mz.$$

Parameter	Exact terminating decimal
k_0	`-0.188363675849724267626172393485401490549571730609919067601324`
d_0	`0.134296165791329499749307346482618861888703361984648841084573`
M_{11}	`-0.000002506480630379097223200313500735928463969728898916674142`
M_{12}	`0.000013485064260217804393465947674708682238388341614402040 65`
M_{21}	`-0.000060574084332297775475288805065049794615157722929526738 56`
M_{22}	`-0.000005283844709378363831746235549426599375331183623652997418`

Table 1: Rational data for the finite certificate.

On the complex parameter polydisc $|k - k_0|, |d - d_0| \le .01$, use the Banach algebra of power series with norm $\sum_i |a_i| 3^i$. On the ball of higher-order terms of radius .2,

$$\|kr + a\| < 3(.198364) + .2 < .796, \qquad |1 + d| > 1.12429.$$

Thus the reciprocal has norm less than 3.1. The denominator in (3.6) is at least 243, so the coefficient map has image norm at most $9(3.1+1)/243 < .2$ and Lipschitz constant at most $9(3.1)^2/243 < 1$. Banach's theorem gives an analytic solution, holomorphic in both parameters. In fact

$$\sum_{i\ge 1} |a_i| 3^i < .75, \qquad a_1 = k. \tag{3.7}$$

This proves positivity for real $0 \le s \le 1$. The jet $(p, p', p'')(1)$ is bounded by 2 in the parameter polydisc, by termwise differentiation and (3.7). Truncating at degree 100 introduces a jet error smaller than

$$.75\,\frac{9}{4}\,101^2\,\frac{3^{-101}}{1-.34} < 10^{-42}.$$

The same Cauchy estimates in the two parameter variables control the first derivatives and the nonlinear remainder in the small square. In the z coordinates the resulting initial error is at most $10^{-38} + .001R^2$; indeed the normalized sum of parameter displacements is at most $.00014R/.01$, and $2u^2/(1-u) < .001R^2$ for this value of u.

3.2 Exact finite-interval enclosure

We state the finite computation with the analytic enclosure that makes it a certificate. All decimal bounds in this subsection are rationals. Intervals have endpoints in $2^{-384}\mathbb{Z}$ and every arithmetic operation rounds outwards. The blocks from $s = 1$ to $s = 56$ are

interval	block length	number
$[1, 2]$	$1/64$	64
$[2, 4]$	$1/32$	64
$[4, 8]$	$1/16$	64
$[8, 56]$	$1/8$	384.

Within each block set $\theta = h(s - s_c)$, where h is the reciprocal block length. Take eight steps of θ-length $1/8$, using Taylor degree 58. In the scaled coordinate the equation is

$$p_{\theta\theta\theta} + \frac{21}{hs_c+\theta}p_{\theta\theta} + \frac{B}{(hs_c+\theta)^2}p_\theta + \frac{C}{(hs_c+\theta)^3}(p-d) = h^{-3}((1+p)^{-1}-1).$$

Taylor coefficients follow by multiplying by $(hs_c+\theta)^3$ and equating powers. If $z = 1+p = \sum z_i\theta^i$, the reciprocal coefficients are $b_0 = z_0^{-1}$ and $b_i = -z_0^{-1}\sum_{j=1}^{i} z_j b_{i-j}$. Differentiating these same recurrences computes the four variational columns, for the three initial jet coordinates and the constant parameter d.

At each step the interval checks give

$$-.923 < p < .4, \quad |p_\theta| < .04, \quad |p_{\theta\theta}| < .01, \quad \text{each variational component has modulus } < 6.$$

Here $h \ge 8$ and $hs_c \ge 64$. On $|\theta| \le 1/2$, bootstrap $|p - p(0)| \le .03$, so $|1+p| > .047$. With $L = 63.5$, the equation, integrated twice and estimated by Gronwall, gives

$$|p_{\theta\theta}| \le \frac{e^{21/(2L)}\left[.01 + (B(.04)/L^2 + |C|(1.5)/L^3 + (.953)/(512\cdot.047))/2\right]}{1 - e^{21/(2L)}B/(4L^2)} < .04.$$

It follows that $|p - p(0)| \le .04/2 + .04/8 < .03$, closing the bootstrap. Analytic ODE continuation proves existence throughout that disk. The variational matrix has norm at most

$$J = 21/L + B/L^2 + 2|C|/L^3 + 1/(512\cdot.047^2) < 1.24.$$

Consequently its components are bounded by $6e^{J/2} < 32$. The state components have the same bound. Cauchy's estimate gives, after up to two differentiations, a remainder at each substep no larger than $48 \cdot 4^{-57}$. On a real substep the change in p is smaller than $.04/8 + .04/128 < .006$.

At each block endpoint replace the enclosed state and variational matrix by their rational midpoints. The enclosure radius in physical jet coordinates, and that of every matrix entry, are each less than 10^{-24}. Denote the midpoint transfer matrices by $\widetilde{T}_j$, their consecutive products by $\widetilde{T}_{i,j}$, and the positive block margins for $1+p$ by m_j. Let Z be the derivative of the initial extended jet with respect to z. Exact arithmetic yields

$$\begin{gathered} m_j > .07, \qquad \max_{i,j} \|\widetilde{T}_{i,j}\|_\infty < 260000, \\ \max_i \sum_{j<i} \|\widetilde{T}^{(3)}_{i,j+1}\|_\infty < 39000000, \quad \max_i \sum_{j<i} \ell_j m_j^{-3} \|\widetilde{T}^{(3)}_{i,j+1}\|_\infty < 127000000, \\ \|Z\|_\infty < .00008, \qquad \max_i \|\widetilde{T}_{i,0} Z\|_\infty < 2.69. \end{gathered} \tag{3.8}$$

Here (3) means the first three rows and columns, and ℓ_j is the block length. All norms are maximum row-sum norms. These assertions are recomputed by the accompanying program `calabi7_verify.py`; it reconstructs the interval integrations from the rational data, without using a previously saved trajectory.

For clarity, the passage from finite arithmetic to exact solutions is as follows. The error in any consecutive transfer product is at most

$$260000 \left[(1 + 4 \cdot 10^{-24} \cdot 260000)^{576} - 1 \right] < 10^{-8}.$$

Thus the two sums in (3.8) for exact transfers are bounded by 40000000 and 130000000. The nonlinear remainder of the reciprocal term is exactly $(\Delta p)^2/((1+p_{\mathrm{ref}})^2(1+p))$ in absolute value. Bootstrap $|\Delta p| \le 10R$. Its forcing is then at most $1.001(10R)^2/m_j^3$. Partial transfers within one block have the needed first-row and third-column bounds less than 3: the scaled variational matrix has norm at most

$$J_r = \max\{1, 21/64 + B/64^2 + 2|C|/64^3 + 1/(512 \cdot .07^2)\}, \qquad e^{J_r} < 3.$$

Returning to the physical first row or third column only inserts factors h^{-j} or h^{i-2} of size at most one. Variation of constants therefore bounds the endpoint error by

$$E \le 40000000 \cdot 10^{-24} + 260001(10^{-38} + .001R^2) + 3 \cdot 130000000 \cdot 1.001(10R)^2 < .04R. \tag{3.9}$$

Within each block, $3(2.69R + E) + 3 \cdot 1.001(10R)^2/.07^3 < 10R$. This improves the bootstrap, preserves positivity, and proves existence of the entire exact two-parameter family up to 56. There is no reliance on positivity only at numerical sample points.

3.3 A genuine half-line solution

Let $S_0 = 56$. Define $p_a(s;d) = \sum_{j=1}^{6} d_j(d) s^{-3j}$ by successive cancellation in (3.5). More explicitly, with $a_0 = 0$, $b_0 = 1$, $a_1 = Cd$, $b_1 = -a_1$, and

$$L(v) = v(v-1)(v-2) + 21v(v-1) + Bv + C,$$

put for $j \ge 2$

$$a_j = -L(-3(j-1))a_{j-1} - \sum_{k=1}^{j-1} a_k b_{j-k}, \qquad b_j = -a_j - \sum_{k=1}^{j-1} a_k b_{j-k}, \qquad d_j = a_j.$$

All are rational polynomials in d. Multiplying the residual by $1+p_a$ produces a finite polynomial in s^{-3}, whose coefficients through degree six vanish exactly. Rational interval evaluation on $.13429 \le d \le .13430$ gives

$$|p_a| < .00011, \qquad |\mathcal{E}_d(p_a)| < 4 \cdot 10^{-22}, \qquad \|\partial_d(p_a, p_a', p_a'')(56)\|_\infty < .001. \tag{3.10}$$

For the last bound, the derivative multipliers $3j/56$ and $3j(3j+1)/56^2$ are at most one for $1 \le j \le 6$, so the bound for the first entry controls all three.

Freeze the coefficient ranges at their midpoints:

$$D_0(\lambda) = \lambda^3 + \frac{21}{112}\lambda^2 + \frac{337}{8 \cdot 56^2}\lambda + 1 - \frac{567}{8 \cdot 56^3}.$$

Its roots are $\lambda_0 = -r_0$ and $\lambda_{1,2} = u \pm iv$, with

$$1.061676856304 < r_0 < 1.061676856305,$$
$$.866302348759 < v < .866302348762, \qquad u = (r_0 - 21/112)/2 > 0.$$

The derivative of this cubic is strictly positive on the real line; the real-root bounds follow by sign evaluation. Vieta's equations give the displayed bounds for u, v. Set $v_i = (1, \lambda_i, \lambda_i^2)^t$ and $b_i = D_0'(\lambda_i)^{-1}$. Decompose real jets as

$$(V, V', V'')^t = q v_0 + \xi v_1 + \bar{\xi} v_2.$$

The projection to $(q, \Re\xi, \Im\xi)$ has row norms less than $.94, .66, .66$. Its componentwise distance from

$$\begin{pmatrix} .314179536995 & -.291705661194 & .333691814294 \\ .342910231503 & .145852830597 & -.166845907147 \\ -.019504134458 & -.324829825596 & -.288655402867 \end{pmatrix}$$

is less than 10^{-10}; these are rational root-interval checks.

For a real source $j \in C_b[S_0, \infty)$ the bounded jet with specified stable coordinate q is

$$\begin{aligned} \mathcal{V}(s) = {} & q e^{\lambda_0(s-S_0)} v_0 + b_0 v_0 \int_{S_0}^{s} e^{\lambda_0(s-u)} j(u)\,\mathrm{d}u \\ & - \sum_{i=1}^{2} b_i v_i \int_{s}^{\infty} e^{\lambda_i(s-u)} j(u)\,\mathrm{d}u. \end{aligned} \tag{3.11}$$

Partial fractions give the companion-system identity $D_0(\partial_s)V = j$. Its three convolution bounds are less than 1.9, since $\sum_i |\lambda_i|^a/(|\Re\lambda_i||D_0'(\lambda_i)|) < 1.9$ for $a = 0, 1, 2$. The unstable initial coordinate satisfies $|\xi| < .8\|j\|_\infty$.

Put $N(p) = p^2/(1+p)$. For $p = p_a + V$, equation (3.5) is the fixed-point equation

$$\begin{aligned} j = {} & -\mathcal{E}_d(p_a) - (21/s - 21/(2S_0))V'' - (B/s^2 - B/(2S_0^2))V' \\ & - (C/s^3 - C/(2S_0^3))V + N(p_a + V) - N(p_a). \end{aligned}$$

On $|q| \le .16R$, $\|j\|_\infty \le .1R$, one has $|p_a + V| < .001$ and $|N'| < .0021$. The Lipschitz constant of this map is less than

$$1.9\{21/(2S_0) + B/(2S_0^2) + |C|/(2S_0^3) + .0021\} < .39.$$

The coefficient of $|q|$ is less than .23. Therefore its image has norm at most $4 \cdot 10^{-22} + .23(.16R) + .39(.1R) < .1R$. This constructs the half-line solution with the prescribed stable coordinate and $|\xi_{\text{tail}}| < .08R$. The solution depends continuously on q, d. Iteration starting from zero stays in C_0, and (3.11) then gives $p, p', p'' \to 0$.

Combining the finite ledger, (3.9), the projection error, and (3.10), the exact finite shot obeys

$$|q_{\text{shot}}(z)| < .15R, \qquad \|(\Re\xi_{\text{shot}}, \Im\xi_{\text{shot}}) - z\|_\infty < .04R. \tag{3.12}$$

The central projected residual is smaller than $.01R$, and the preconditioned unstable derivative differs from the identity by less than 10^{-6}; all allowances are included by the checker. For each z choose the tail with $q = q_{\text{shot}}(z)$ and the same d. The continuous map

$$z \longmapsto z - (\Re(\xi_{\text{shot}} - \xi_{\text{tail}}), \Im(\xi_{\text{shot}} - \xi_{\text{tail}}))$$

has image in the square of radius $.12R < R$. Brouwer's theorem makes all three jet coordinates agree. ODE uniqueness joins the finite shot to the tail. Positivity on the finite interval and $|p| < .001$ on the tail imply the global positive lower and upper bounds. Since $r = \sqrt{27}\, y/(4c)$, $G'(0) = \sqrt{27}k/4 < 0$.

3.4 Differentiated expansion at infinity

The remaining step upgrades $p, p', p'' \to 0$ to the expansion used in the parabolic argument. Successively cancel powers to form $p_J = \sum_{j=1}^{J} d_j s^{-3j}$. Its residual e_J satisfies $\partial_s^\ell e_J = O(s^{-3(J+1)-\ell})$. Set $W = p - p_J$ and $N_0 = 3(J+1)$. Subtracting equations gives

$$W''' + W = -e_J - a_2 W'' - a_1 W' - a_0 W,$$

where $a_2 = 21/s$, $a_1 = B/s^2$, and $a_0 = C/s^3 + (1+p)^{-1}(1+p_J)^{-1} - 1$. All tend to zero; they and their derivatives are bounded on a sufficiently late half-line. Boundedness of all derivatives of p follows inductively by differentiating (3.5), starting with $p, p', p'' \to 0$.

The Green operator for $\partial_s^3 + 1$, with stable data at S and unstable data at infinity, is bounded on both the bounded-jet space and the space with norm $\sup_{s \ge S} s^{N_0}(|W| + |W'| + |W''|)$. For the latter assertion split each exponential convolution into $|s-u| \le s/2$ and its complement; polynomial weight ratios are bounded in the first part, and exponential decay dominates them in the second. For fixed N_0 these operator bounds are uniform for all sufficiently large S. Take S so large that multiplication by (a_2, a_1, a_0) followed by either Green operator has norm less than one. Choose the stable coordinate equal to that of the actual W at S. The affine source equation has a unique solution in each space. The weighted solution is bounded, so uniqueness identifies it with the actual remainder. This proves $W, W', W'' = O(s^{-N_0})$. The differential equation and its successive derivatives give $W^{(\ell)} = O(s^{-N_0})$ for each fixed ℓ. To obtain the sharper $O(s^{-N_0-\ell})$, apply this result to $p_{J'}$ with $3(J'-J) \ge \ell$, and subtract the finite intervening polynomial. Its differentiated leading term has exactly the desired order. The change of variables (3.4) proves (3.3). This completes Theorem 3.1.

4 Parabolic norms and uniform local estimates

The correction to the profile must satisfy the lower-order bound in Lemma 2.3. To encode that bound together with its spatial derivatives, put

$$\rho = \sqrt{-t}, \qquad r = \sqrt{x}, \qquad \sigma = (r^2 + \rho)^{1/2}, \qquad p_0 = -2 + 2\delta, \qquad 0 < \delta < 1.$$

At the concentrating point, a bound $|V| \le \varepsilon\sigma^{p_0}$ gives $|V(t,0)| \le \varepsilon\rho^{-1+\delta}$, smaller than the profile's ρ^{-1} term. On the sphere, $v_* \le r^2 \le \sigma^2$, so it also gives $|v_*V| \le \varepsilon\sigma^{2\delta}$. Since σ is bounded on $\Sigma\times[-T,0)$ for fixed T, a small weighted norm will preserve positivity of the multiplier $1+v_*(F+V)$. Outside the shrinking core, $r^2 \ge \rho$, the scale σ is comparable to r and the same weight records spatial decay. The particular value of δ will be chosen in Theorem 5.1.

The norms below measure four spatial derivatives and one time derivative in units (σ,σ^4). We also need estimates for an outer forward problem with zero initial data, uniform as the elapsed time tends to zero. For this purpose we distinguish the full parabolic norm from the norm that omits the first time derivative.

4.1 Norms and scales

On a product $D = U \times I \subset \mathbb{R}^d \times \mathbb{R}$, put

$$d_{\mathrm{p}}((z,t),(w,s)) = |z-w| + |t-s|^{1/4}.$$

Write

$$[g]_{\alpha;D} = \sup_{P\neq Q\in D} \frac{|g(P)-g(Q)|}{d_{\mathrm{p}}(P,Q)^\alpha}, \quad |g|_{\alpha;D} = \|g\|_{\infty;D} + [g]_{\alpha;D}.$$

The joint seminorm is equivalent, with constants depending only on α, to the sum of the spatial α and temporal $\alpha/4$ seminorms. We use the following explicit version of the fourth-order parabolic norm:

$$\mathcal{N}_\alpha(u;D) = \sum_{j=0}^{4} |D_z^j u|_{\alpha;D} + |u_t|_{\alpha;D}, \qquad \mathcal{N}'_\alpha(u;D) = \sum_{j=0}^{4} |D_z^j u|_{\alpha;D}. \tag{4.1}$$

Here and below norms of arrays mean sums of component norms. Thus $\mathcal{N}'_\alpha$ still includes temporal Hölder seminorms of all four spatial derivatives.

Definition 4.1 (Weighted local spaces)**.** Let M be either $\mathbb{R}^m$ or the fixed sphere Σ, let I be a time interval, and let $s(P) > 0$ be a scale on $M \times I$. Fix a sufficiently small number $a_0 > 0$. In Euclidean coordinates, or normal coordinates on Σ, centered at $P = (z_0, t_0)$, write $s_0 = s(P)$ and set

$$\Psi_P(y,\theta) = (\exp_{z_0}(s_0 y), t_0 + s_0^4\theta), \qquad D_P = B_{a_0} \times \big((-a_0^4, a_0^4) \cap s_0^{-4}(I - t_0)\big).$$

In the Euclidean case the exponential map means translation. The one-sided endpoints belonging to I are included. For a fixed weight p, define

$$\|u\|_{X_{\alpha,p}(s)} = \sup_P \mathcal{N}_\alpha(s_0^{-p} u\circ\Psi_P; D_P), \tag{4.2}$$

$$\|u\|_{X'_{\alpha,p}(s)} = \sup_P \mathcal{N}'_\alpha(s_0^{-p} u\circ\Psi_P; D_P), \tag{4.3}$$

$$\|f\|_{Y_{\alpha,p}(s)} = \sup_P |s_0^{4-p} f\circ\Psi_P|_{\alpha;D_P}. \tag{4.4}$$

The derivatives in these formulas are derivatives of the displayed pullbacks. On the sphere s is bounded above so that these charts, and their fixed enlargements used below, lie within the injectivity radius. Restriction to radial functions on $\mathbb{R}^m$, or to invariant functions on Σ, gives closed subspaces with these same norms.

Unless specified otherwise, X, X', Y mean the spaces (4.2)–(4.4) with $s = \sigma$, $p = p_0$, and the fixed exponent $0 < \alpha < 1$. Subscripts α' indicate replacement of α by α', with $\alpha < \alpha' < 1$.

These are local weighted norms; no Hölder quotient between distant boxes is imposed. Their completeness follows by convergence of the functions and indicated derivatives on every compact subset, followed by taking the supremum in the defining inequalities. The derivative relations pass to the limit distributionally. In particular X and X' are Banach spaces and $X \hookrightarrow X'$ has norm at most one.

The radius function r is Lipschitz on both spaces. The elementary inequalities for Euclidean norms and fourth roots give

$$|\sigma(z,t) - \sigma(z_0,t_0)| \le |r(z) - r(z_0)| + |t - t_0|^{1/4}.$$

Consequently σ is comparable to its center value on every box in Definition 4.1, including boxes truncated at $t = 0$. Changing a_0 to another fixed sufficiently small radius gives equivalent norms: cover the enlarged box by finitely many smaller comparable-scale boxes, and split a Hölder quotient whose points are not in a common small box using the supremum norm. The same argument applies to $s_\beta = \max(\beta, r)$, whose lack of smoothness at $r = \beta$ is harmless: the scale is frozen at each center, never differentiated. Products satisfy the usual local Hölder inequality $|fg|_\alpha \le 2|f|_\alpha |g|_\alpha$, with their weights added.

4.2 Interpolation without a lower bound for the time length

We first isolate the interpolation used in localization. For fixed nested balls $B_1 \Subset B_2$ and any interval I of length at most one, let

$$U = \|u\|_{\infty;B_2\times I}, \qquad M = U + \|D_z^4 u\|_{\infty;B_2\times I} + \|u_t\|_{\infty;B_2\times I}.$$

Then, for $0 \le j \le 3$,

$$|D_z^j u|_{\alpha;B_1\times I} \le C U^{1-(j+\alpha)/4} M^{(j+\alpha)/4} \le \varepsilon M + C_\varepsilon U. \tag{4.5}$$

All constants are independent of $|I|$. Spatial interpolation on nested balls gives the spatial assertion, including the supremum. For the temporal assertion apply that same interpolation to $w = u(\cdot,t) - u(\cdot,s)$. With $b = |t - s|$,

$$\|w\|_\infty \le \min(2U, bM), \qquad \|D^4 w\|_\infty \le 2M,$$

and hence

$$\|D^j w\|_{\infty;B_1} \le C \min(2U, bM)^{1-j/4} M^{j/4}.$$

Divide by $b^{\alpha/4}$ and split at $b = U/M$; both resulting bounds give (4.5). Restricting the available values of b can only decrease their supremum. The spatial interpolation inequalities themselves follow, for example, by subtracting a Taylor polynomial in a ball of radius q and minimizing in q; fixed-radius remainders account for the U term when the minimizer exceeds the gap between the balls. The same argument at order four gives

$$\|D^4 u\|_{\infty;B_1\times I} \le \varepsilon [D^4 u]_{\alpha;B_2\times I} + C_\varepsilon U. \tag{4.6}$$

Here it suffices to use the spatial part of the seminorm on the right. Indeed, convolution on each spatial slice with a mollifier of radius q gives $\|D^4 u\|_{B_1} \le C q^\alpha [D^4 u]_{\alpha;B_2} + C q^{-4} U$; choose q below the spatial gap and so that $C q^\alpha \le \varepsilon$. Neither inequality uses a temporal seminorm of u_t.

4.3 Local fourth-order estimates

Consider in a coordinate cylinder the real scalar operator

$$\mathcal{L}u = u_t + \sum_{|\nu|\le 4} a_\nu(z,t) D^\nu u, \qquad \sum_{|\nu|=4} a_\nu(z,t)\xi^\nu \ge \lambda|\xi|^4. \tag{4.7}$$

Assume $\lambda > 0$ and $\sum_{|\nu|\le 4} |a_\nu|_\alpha \le K$. These assumptions are always imposed in the coordinates and units of the estimate. On a fixed smooth manifold they include the coefficients arising from the coordinate expression of the operator. The leading symbols used later are positive multiples of squared positive definite quadratic forms, and satisfy (4.7).

Lemma 4.2 (Local Schauder estimates)**.** *Under these assumptions the following estimates hold.*

(i) *If $Q_R = B_R \times (-R^4, 0]$ and $\mathcal{L}u = f$ on Q_2, then*

$$\mathcal{N}_\alpha(u; Q_1) \le C(|f|_{\alpha;Q_2} + \|u\|_{\infty;Q_2}). \tag{4.8}$$

(ii) *If $0 < h \le 1$, $\mathcal{L}u = f$ on $B_2 \times [0,h]$, and $u(\cdot,0) = 0$, then*

$$\mathcal{N}_\alpha(u; B_1 \times [0,h]) \le C(|f|_{\alpha;B_2\times[0,h]} + \|u\|_{\infty;B_2\times[0,h]}). \tag{4.9}$$

Here C depends only on d, α, λ, K and the fixed spatial gaps, and in (ii) *is independent of h.*

These are a priori estimates for classical solutions in the indicated spaces; smooth approximation gives the same conclusions for limits of such solutions. The usual interior regularity theorem also applies: with smooth coefficients, distributional solutions in L^2_{loc} and Hölder sources are locally in the space (4.1)*, and are smooth when the source is smooth. The estimates are from preceding times and require no terminal data.*

Proof. We give the kernel and localization argument, particularly its uniformity in (ii). For a frozen homogeneous principal operator A_0, with symbol $p(\xi) \ge \lambda|\xi|^4$, its heat kernel is

$$k_b(z) = (2\pi)^{-d} \int_{\mathbb{R}^d} e^{iz\cdot\xi - bp(\xi)}\,\mathrm{d}\xi, \qquad b > 0.$$

Fourier differentiation and scaling give, uniformly for the allowed symbols and every N, j, ℓ,

$$|D^j \partial_b^\ell k_b(z)| \le C_{N,j,\ell} b^{-(d+j)/4-\ell}(1 + |z|b^{-1/4})^{-N}. \tag{4.10}$$

Moreover $\int k_b = 1$ and $\int D^j k_b = 0$ for $j > 0$. For $u(t) = \int_0^t k_{t-s} * f(s)\,\mathrm{d}s$ and $j = 4$, subtract $f(s,z)$ inside the convolution. Writing $F = \sup_s [f(s)]_{\alpha;\mathbb{R}^d}$,

$$\|D^4 k_b * f(s)\|_\infty \le CFb^{-1+\alpha/4}, \qquad \|\partial_b D^4 k_b * f(s)\|_\infty \le CFb^{-2+\alpha/4}. \tag{4.11}$$

This proves $\|D^4 u(t)\|_\infty \le CFt^{\alpha/4}$. For a spatial increment of size q, split the lag integral at q^4. The short-lag part is bounded by $CF\int_0^{q^4} b^{-1+\alpha/4}\,\mathrm{d}b$. For the long-lag part use the mean value theorem and $\|D^5 k_b * f(s)\|_\infty \le CFb^{(\alpha-5)/4}$. Its bound is $CFq\int_{q^4}^h b^{(\alpha-5)/4}\,\mathrm{d}b$. Both are at most CFq^α, with an empty integral omitted.

For a temporal increment $e = t - s > 0$, keep the source time unchanged:

$$\begin{aligned} D^4 u(t) - D^4 u(s) = &\int_s^t D^4 k_{t-r} * f(r)\,\mathrm{d}r \\ &+ \int_0^s (D^4 k_{t-r} - D^4 k_{s-r}) * f(r)\,\mathrm{d}r. \end{aligned}$$

The first integral and the part of the second where $s - r < e$ are bounded by $CFe^{\alpha/4}$. On the remaining part, (4.11) bounds the difference by $CFe(s-r)^{-2+\alpha/4}$, and

$$e\int_e^s b^{-2+\alpha/4}\,\mathrm{d}b \le Ce^{\alpha/4}.$$

This proves the full parabolic α bound for D^4u, down to the initial face, without any requirement that $f(0)$ vanish. The bounds for u and lower derivatives follow from the kernel integrals or (4.5); finally $u_t = f - A_0u$ supplies the full parabolic norm of u_t. We have proved the constant-coefficient zero-data estimate on every slab of length at most one, with one constant.

Here are the localization details needed to preserve that constant. We apply the fourth-derivative frozen estimate in the original local units before localization; its constant is independent of the localization radius. In a spatial ball of radius q and a time window of length at most q^4, freeze the principal coefficients at one point. Use a spatial cutoff supported in that ball. For windows meeting the initial face use no lower-time cutoff, so the localized function still has zero initial value. For interior windows insert a time cutoff which vanishes at the bottom and equals one on the smaller window; its derivatives are bounded by fixed powers of q^{-1}. The constant-coefficient estimate then applies in both cases. The localized forcing consists of the original source, terms of spatial order at most three, a time-cutoff multiple of u, and $(a^0_\nu - a_\nu)D^\nu u$ for $|\nu| = 4$. The last terms obey

$$|(a^0_\nu - a_\nu)D^\nu u|_\alpha \le CKq^\alpha |D^4u|_\alpha + CK\|D^4u\|_\infty, \tag{4.12}$$

with fixed cutoff factors understood. First make q small enough to absorb the first coefficient. Apply (4.6) to the second term. For the lower-order terms apply (4.5), and use the equation only in the form

$$\|u_t\|_\infty \le \|f\|_\infty + K\sum_{j=0}^{4}\|D^ju\|_\infty. \tag{4.13}$$

Thus no unknown temporal Hölder seminorm of u_t is used to absorb a temporal seminorm of D^4u.

For completeness, the resulting nested-domain absorption can be expressed as follows. Let $H(r)$ be the supremum and parabolic α norm of D^4u on the spatial ball of radius r, with the given time interval in the initial case; in the interior case also increase the preceding-time window with r. For $1 \le r < R \le 3/2$, localization, interpolation, and covering give, for any fixed $\vartheta > 0$,

$$H(r) \le \vartheta H(R) + C_\vartheta (R-r)^{-k}(|f|_\alpha + \|u\|_\infty).$$

Here k is fixed; all losses are powers of spatial cutoff radii, and none is a power of h^{-1}. For Hölder pairs not sharing a small localization window, the quotient is bounded by the corresponding inverse power of that window's radius times $\|D^4u\|_\infty$, which is handled by (4.6). Iterate along radii increasing geometrically to $3/2$, choosing $\vartheta 2^k < 1$. The last term tends to zero since the solution norm is finite on that compact cylinder. This proves the estimate for D^4u. Equations (4.5) and (4.13) then give the lower derivatives, and the equation in full Hölder norm gives $|u_t|_\alpha$. This proves (i) and (ii).

For the distributional regularity assertion we use the standard local regularity theorem for smooth strongly parabolic operators. Its hypotheses hold here: the principal symbol is $i\tau + a_4(x,t,\xi)$ with $a_4 \ge \lambda|\xi|^4$, and the remaining symbol has spatial degree at most three. On compact coordinate sets the full symbol is bounded below by $c(|\tau| + |\xi|^4)$ at high frequency, and each frequency derivative improves this bound by at least one fourth of a power of $1 + |\tau| + |\xi|$. Localization and the parabolic parametrix therefore give interior regularity for distributional solutions. One can first subtract a local $C^{4+\alpha,1+\alpha/4}$ particular solution, obtained from the smooth Cauchy theorem, and apply this

regularity to the homogeneous difference. This use requires no estimate uniform in a degenerating time or spatial scale; the uniform a priori estimates were proved above. □

Lemma 4.3 (Interior estimate with an L^2 input)**.** *Let $U_1 \Subset U_2$ be fixed bounded spatial domains and $b_0 < b_1 < b$, with $b - b_1$ bounded above. Suppose that the coefficient assumptions of Lemma 4.2 hold on $U_2 \times (b_0, b)$ and that $\mathcal{L}u = f$ there. Then*

$$\mathcal{N}_\alpha(u; U_1 \times [b_1, b)) \le C\big(|f|_{\alpha; U_2 \times (b_0,b)} + \|u\|_{L^2(U_2 \times (b_0,b))}\big). \tag{4.14}$$

The constant depends on the spatial gap, $b_1 - b_0$, the upper bound for $b - b_1$, and the coefficient constants, but requires no terminal trace. The estimate is unchanged if the upper face is included and the solution is defined there. In particular it holds on fixed nested annuli through an open upper time endpoint.

Proof. We recall an interpolation proof of the replacement of the supremum term. On three nested interior cylinders, with a gap of parabolic size a, the estimate already proved bounds $\|D_z u\|_\infty$ and $\|u_t\|_\infty$ on the middle cylinder by $Ca^{-k}(\|u\|_{\infty;\text{outer}} + |f|_\alpha)$. Average u on a preceding cylinder of radius $q < a/4$ ending at the point being estimated. Its volume is a fixed multiple of q^{d+4}, so

$$|u(z,t)| \le Cq^{-(d+4)/2}\|u\|_{L^2(\text{outer})} + C(q + q^4)a^{-k}(\|u\|_{\infty;\text{outer}} + |f|_\alpha).$$

Choose q to be a sufficiently small fixed multiple of a suitable power of a. The last supremum coefficient can be made any prescribed $\vartheta > 0$, at the cost of another fixed inverse power of a on the other terms. Iteration over geometrically decreasing gaps, with ϑ smaller than the corresponding geometric growth factor, bounds the inner supremum by $C(\|u\|_{L^2} + |f|_\alpha)$. Lemma 4.2 on one further nested cylinder proves (4.14); finite covering handles U_1 and the time interval. All averaging cylinders use preceding times. First carry this out up to any closed upper face $b' < b$ on which the local norms are finite, then let $b' \uparrow b$. The constants use only the fixed lower time buffer. Pairs of points approaching b are included by this limiting argument. □

4.4 The outer forward problem

The following formulation permits the localization scale to tend to zero. Fix a compact smooth manifold without boundary and a time-independent positive Lipschitz scale $s(z)$. Assume that on balls of radius $4a_0 s(z)$ the scale is comparable to its center value, with uniform comparison constants and uniformly bounded coordinate changes in scaled units. We impose the coefficient assumptions of (4.7) uniformly after scaling space by $s(z)$ and time by $s(z)^4$; an original coefficient of spatial order j is thus multiplied by $s(z)^{4-j}$. The ellipticity and Hölder constants in these units are denoted again by λ, K. These hypotheses hold for $s = s_\beta$ in the later application.

Lemma 4.4 (Uniform zero-data forward estimate)**.** *Fix $p \in \mathbb{R}$, $0 < \alpha < 1$, and the preceding scale and coefficient constants. There exist $\eta_0 > 0$ and $C < \infty$, independent of* $\inf s$*, with the following property. If*

$$0 < T \le \eta_0(\inf s)^4, \qquad \mathcal{L}u = f \quad \text{on } M \times [0,T], \qquad u(\cdot, 0) = 0,$$

then, with $\eta = T/(\inf s)^4$,

$$\|u\|_{X_{\alpha,p}(s)} \le C\|f\|_{Y_{\alpha,p}(s)}, \qquad \sup_{z,t} s(z)^{-p}|u(z,t)| \le C\eta\|f\|_{Y_{\alpha,p}(s)}. \tag{4.15}$$

For each fixed positive scale and smooth coefficients the zero-data Cauchy problem has a unique solution with this regularity for $f \in Y_{\alpha,p}(s)$. The same estimates hold on every interval $[0, T')$, $T' \le T$, on which the coefficients and source obey the stated bounds.

Proof. Choose $\eta_0 < a_0^4$. Every defining box meets the initial face, and its full time interval has length at most η_0 in center-scale units. Apply (4.9) on a fixed spatial enlargement of each such box, rescaling its fixed spatial radius if necessary. Scale comparability and the covering observation after Definition 4.1 give

$$\|u\|_{X_{\alpha,p}(s)} \le C(\|f\|_{Y_{\alpha,p}(s)} + S), \qquad S = \sup_{z,t} s(z)^{-p}|u(z,t)|.$$

The pointwise time-derivative bound in this norm and zero initial data imply

$$S \le \sup_z \frac{T}{s(z)^4}\|u\|_{X_{\alpha,p}(s)} \le \eta\|u\|_{X_{\alpha,p}(s)}.$$

Decrease η_0 so that $C\eta_0 \le 1/2$, and absorb. This proves both estimates in (4.15) with constants independent of the number or size of the spatial boxes.

Fixed-scale existence and uniqueness follow from the smooth strongly parabolic Cauchy theorem on a compact manifold without boundary, applied to the scalar fourth-order operator $\mathcal{L} = \partial_t + A(t)$ with zero initial data. Its solution is $C^{4+\alpha,1+\alpha/4}$ on the closed time slab, without requiring $f(\cdot,0) = 0$; the equation gives $u_t(\cdot,0) = f(\cdot,0)$. Constants in this fixed-scale existence argument need not be uniform in $\inf s$; uniformity of the asserted estimates has already been established locally. Truncation before an open terminal face and passage to the limit prove the last statement. □

Lemma 4.5 (Small spatial derivatives on a short interval)**.** *Under the hypotheses of Lemma 4.4, the local scaled parabolic C^α norms of the spatial derivatives of order $0 \le j \le 3$, with weight $p - j$, satisfy*

$$\sup_P |D_y^j(s_0^{-p}u \circ \Psi_P)|_{\alpha;D_P} \le C\eta^{(4-j-\alpha)/4}\|f\|_{Y_{\alpha,p}(s)}. \tag{4.16}$$

If the coefficient and source bounds also hold at an exponent $\alpha' \in (\alpha, 1)$, then

$$\|u\|_{X'_{\alpha,p}(s)} \le C\eta^{(\alpha'-\alpha)/4}\|f\|_{Y_{\alpha',p}(s)}. \tag{4.17}$$

Constants may depend on α' but remain independent of the scale and elapsed time.

Proof. On a fixed enlarged scaled box, (4.15) gives $U \le C\eta\|f\|_Y$ and $M \le C\|f\|_Y$ in the notation of (4.5). That inequality proves (4.16). In particular, all derivatives entering a commutator of a fourth-order spatial operator with a spatial cutoff have a common sufficient small factor $\eta^{(1-\alpha)/4}$.

For the second claim apply Lemma 4.4 at exponent α', decreasing η_0 if necessary. Every spatial derivative of u through order four has zero initial trace. Indeed the solution and these derivatives are continuous to the initial face, and the trace function $u(\cdot,0)$ is identically zero. For each $g = D_y^j(s_0^{-p}u \circ \Psi_P)$, $0 \le j \le 4$, the temporal $\alpha'/4$ seminorm and the elapsed scaled time therefore give

$$\|g\|_\infty \le C\eta^{\alpha'/4}\|f\|_{Y_{\alpha',p}(s)}, \qquad [g]_{\alpha'} \le C\|f\|_{Y_{\alpha',p}(s)}.$$

On any metric space, splitting distances where the estimates $|g(P) - g(Q)| \le 2\|g\|_\infty$ and $|g(P) - g(Q)| \le [g]_{\alpha'}d(P,Q)^{\alpha'}$ agree proves

$$[g]_\alpha \le C\|g\|_\infty^{1-\alpha/\alpha'}[g]_{\alpha'}^{\alpha/\alpha'}.$$

Apply this with the parabolic metric and take the supremum over boxes. It proves (4.17), including temporal Hölder quotients of D^4u with one point at the initial face. □

Remark 4.6. The exponent gap and omission of u_t in (4.17) are necessary. On a circle the exact zero-data solution

$$u_N(z,t) = N^{-4-a}(1 - e^{-N^4 t})\cos(Nz), \qquad f_N(z) = N^{-a}\cos(Nz), \qquad T_N = N^{-4},$$

of $u_t + \partial_z^4 u = f_N$ has $[\partial_z^4 u_N]_\alpha$ of order $N^{\alpha-a}$ on $[0, T_N]$. This need not tend to zero when $a = \alpha$, whereas $a = \alpha'$ gives precisely the power in (4.17). Moreover $u_t(0) = f_N$. Thus the argument gives smallness of the full spatial norm X', while making no smallness assertion for the time derivative.

5 An ancient inverse for the flat linearization

Fix the profile supplied by Theorem 3.1, and put

$$\rho = \sqrt{-t}, \qquad x = |z|^2, \qquad h(y) = \frac{G(y) - 1}{y}, \qquad F(z,t) = \rho^{-1} h(|z|^2/\rho), \qquad z \in \mathbb{R}^m,$$

where $m = 18$ and $h(0) = G'(0)$. Throughout this section the time domain is the entire interval $(-\infty, 0)$. The flat linearization is

$$\widehat{\mathcal{P}} V = V_t + \frac{A(z,t)}{16}\Delta_z^2 V + B(z,t)V, \quad A = G(|z|^2/\rho)^2, \quad B = 2|z|^2 G(|z|^2/\rho)\widehat{\mathcal{A}}F, \qquad \widehat{\mathcal{A}} = \frac{1}{16}\Delta_z^2. \tag{5.1}$$

The spaces below are the radial versions of Definition 4.1, with scale $\sigma = (|z|^2 + \rho)^{1/2}$.

Theorem 5.1 (Ancient flat right inverse)**.** *There is a number $\delta \in (0,1)$ such that, for every fixed $\alpha \in (0,1)$, the operator* (5.1) *has a bounded real linear right inverse*

$$\widehat{D} : Y \longrightarrow X, \qquad \widehat{\mathcal{P}}\widehat{D} f = f, \qquad \|\widehat{D} f\|_X \le C\|f\|_Y, \qquad p_0 = -2 + 2\delta.$$

Here Y has weight $p_0 - 4$ and parabolic exponent α, and X has weight p_0 and includes the parabolic α norms of all spatial derivatives through order four and of the first time derivative. The solution is radial and classical on $t < 0$. Its initial values are selected by the construction. The constant depends only on the fixed profile, δ, α, and the fixed choices in the local norms.

5.1 Similarity coordinates and coefficient bounds

The flat equation for F in Theorem 3.1 and the radial operator identity of Lemma 2.2 yield

$$\frac{1}{16}\Delta_\zeta^2(h(|\zeta|^2)) = -\frac{G'(y)}{2G(y)^2}, \qquad y = |\zeta|^2, \qquad F_t + G(|z|^2/\rho)^2\widehat{\mathcal{A}}F = 0. \tag{5.2}$$

In particular, with

$$A_f(\zeta) = G(|\zeta|^2)^2, \qquad B_f(\zeta) = \frac{2|\zeta|^2 G(|\zeta|^2)}{16}\Delta_\zeta^2(h(|\zeta|^2)) = -\frac{yG'(y)}{G(y)},$$

one has $A(z,t) = A_f(z/\sqrt{\rho})$ and $B(z,t) = \rho^{-2}B_f(z/\sqrt{\rho})$. These coefficients are smooth at the origin. Positivity and the differentiated asymptotic expansion in Theorem 3.1 imply

$$\inf A_f > 0, \qquad |\partial_\zeta^\nu(A_f - c^2)| + |\partial_\zeta^\nu B_f| \le C_\nu(1 + |\zeta|)^{-4-|\nu|}. \tag{5.3}$$

In particular all derivatives, including any fixed number of applications of $\zeta \cdot \nabla$, are bounded.

Set

$$\tau = -\log(-t), \qquad \zeta = z/\sqrt{\rho}, \qquad V(z,t) = \rho^{-1} v(\zeta,\tau), \qquad f^{\#}(\zeta,\tau) = \rho^{3} f(\sqrt{\rho}\,\zeta, t).$$

Since $\mathrm{d}\tau/\mathrm{d}t = \rho^{-2}$ and $\partial_t \zeta = \zeta/(4\rho^2)$, direct differentiation gives

$$V_t = \rho^{-3}\left(v_\tau + \tfrac{1}{4}\zeta\cdot\nabla v + \tfrac{1}{2}v\right), \qquad \Delta_z^2 V = \rho^{-3}\Delta_\zeta^2 v.$$

Thus (5.1) becomes the autonomous equation

$$v_\tau + \frac{A_f}{16}\Delta^2 v + \frac{1}{4}\zeta\cdot\nabla v + (\tfrac{1}{2} + B_f)v = f^{\#}. \tag{5.4}$$

We shall first solve it in $E = C_{0,\mathrm{rad}}(\mathbb{R}^m;\mathbb{C})$, with the supremum norm. This denotes continuous radial functions tending to zero at spatial infinity; complex scalars are needed only for spectral theory.

5.2 Evolution on continuous functions vanishing at infinity

We give the evolution construction since the unbounded drift in (5.4) and the change in the principal coefficient both matter. On a time interval beginning at τ_0, put

$$u(\eta, s) = v(e^{s/4}\eta, \tau_0 + s), \qquad 0 \le s \le 1.$$

Then

$$u_s + a(\eta,s)\Delta_\eta^2 u + b(\eta,s)u = l(\eta,s), \qquad a = \frac{e^{-s}}{16}A_f(e^{s/4}\eta), \quad b = \tfrac{1}{2} + B_f(e^{s/4}\eta). \tag{5.5}$$

The right side is the corresponding dilation of $f^{\#}$. By (5.3), a, b have bounded derivatives of every order on this slab, and $0 < \kappa \le a \le \kappa^{-1}$ for a fixed κ. The following construction also applies without radial symmetry. It is the scalar whole-space version of the higher-order heat-kernel parametrix, with the relevant construction given below.

Let

$$k_\theta(w) = (2\pi)^{-m}\int_{\mathbb{R}^m} e^{iw\cdot\xi - \theta|\xi|^4}\,\mathrm{d}\xi, \qquad \theta > 0.$$

Its integral is one. Scaling and integration by parts in the Fourier integral show that, uniformly for $\kappa \le a_* \le \kappa^{-1}$,

$$|\partial_{a_*}^{\ell}\partial_w^{\nu} k_{a_* q}(w)| \le C_{N,\ell,\nu}\, q^{-(m+|\nu|)/4}(1 + |w|/q^{1/4})^{-N}, \qquad 0 < q \le 1. \tag{5.6}$$

Indeed after scaling $\xi = q^{-1/4}\xi'$ the multipliers and all their derivatives form a bounded set of Schwartz functions. No sign property of k_θ is used.

Freeze the principal coefficient at the source point and time:

$$K_{s,r}(\eta, y) = k_{a(y,r)(s-r)}(\eta - y), \qquad r < s,$$

and also denote by $K_{s,r}$ the associated integral operator. Applying the left side of (5.5) at the evaluation point gives the residual kernel

$$J_{s,r}(\eta,y) = (a(\eta,s) - a(y,r))\Delta_\eta^2 k_{a(y,r)(s-r)}(\eta - y) + b(\eta,s)K_{s,r}(\eta,y). \tag{5.7}$$

The global Lipschitz bound for a and (5.6) give

$$\|K_{s,r}\|_{C_0\to C_0} \le C, \qquad \|J_{s,r}\|_{C_0\to C_0} \le C\big((s-r)^{-3/4}+1\big) \le C'(s-r)^{-3/4}. \tag{5.8}$$

For the first residual term, the spatial difference $|\eta - y|$ contributes $(s-r)^{1/4}$ against a fourth derivative of size $(s-r)^{-1}$; the temporal difference contributes $(s-r)$. The kernel bounds first show that both operators preserve C_0 for compactly supported continuous inputs, by decay in η, and then for all C_0 inputs by uniform approximation.

These kernels give a strong approximate identity. To see this explicitly, replace $a(y,r)$ by $a(\eta,r)$ in $K_{s,r}$. The operator norm of the difference is $O((s-r)^{1/4})$, by the parameter derivative bound in (5.6). For the replacement kernel, its integral in y is one at each η; uniform continuity of a C_0 function, followed by the uniform kernel tail bound, proves convergence to that function in supremum norm. Consequently

$$K_{s,r}d \longrightarrow d \quad \text{in } C_0 \quad \text{as } s-r \downarrow 0, \tag{5.9}$$

uniformly in the initial time r. This is strong convergence; only the difference between the two frozen kernels was estimated in operator norm.

For $d \in C_0$ and $l \in C([0,1];C_0)$ solve

$$\begin{aligned} j(s) + \int_0^s J_{s,r}j(r)\,\mathrm{d}r &= l(s) - J_{s,0}d, \\ u(s) &= K_{s,0}d + \int_0^s K_{s,r}j(r)\,\mathrm{d}r. \end{aligned} \tag{5.10}$$

All integrals are Bochner integrals. The inverse of the first equation is its Volterra series. For completeness, if $H(s) = Cs^{-3/4}$, its k-fold convolution on the positive half-line is

$$H^{*k}(s) = \frac{C^k\Gamma(1/4)^k}{\Gamma(k/4)}s^{k/4-1} \quad (k \ge 1).$$

This identity follows by the beta integral and induction. Since the right side of the first equation in (5.10) is bounded by $Cs^{-3/4}(\|d\|_\infty + \|l\|_{C([0,1];C_0)})$, the same formula and the growth of the gamma function prove convergence of the series and the estimates

$$\|j(s)\|_\infty \le Cs^{-3/4}(\|d\|_\infty + \|l\|_{C([0,1];C_0)}), \qquad \sup_{0\le s\le 1}\|u(s)\|_\infty \le C(\|d\|_\infty + \|l\|_{C([0,1];C_0)}). \tag{5.11}$$

The integral in the formula for u tends to zero as $O(s^{1/4})$. Together with (5.9), this proves $u \in C([0,1];C_0)$ and $u(0) = d$. Continuity at positive times follows by dominated convergence, splitting off a short interval adjacent to the diagonal.

The kernel identity (5.7) proves that u solves (5.5) distributionally: apply the equation to the second line of (5.10), obtaining $J_{s,0}d + j(s) + \int_0^s J_{s,r}j(r)\,\mathrm{d}r = l(s)$. This calculation can be made first with a positive separation from the diagonal and smooth compactly supported tests; the diagonal term converges to j by (5.9), and (5.8) justifies the remaining integrals. Lemma 4.2 and interior regularity then give smoothness for positive times in the homogeneous case, and classical local regularity for locally parabolic Hölder sources.

We also record uniqueness in the class just constructed. If u is the difference of two such solutions, it solves the homogeneous equation with zero C_0 initial value. Let $W(\eta) = \exp(-\sqrt{1+|\eta|^2})$. Its derivatives are bounded by constant multiples of W. For $s \ge \varepsilon > 0$, interior regularity supplies

globally bounded spatial derivatives of u. Integration by parts, first with compact spatial cutoffs, gives

$$\Re \int W \overline{u}\, a\Delta^2 u = \Re \int \Delta(aW\overline{u})\Delta u \geq \frac{\kappa}{2}\|\Delta u\|^2_{L^2(W)} - C\|u\|^2_{L^2(W)}.$$

Here the mixed terms are absorbed using

$$\|\nabla u\|^2_{L^2(W)} \leq \epsilon \|\Delta u\|^2_{L^2(W)} + C_\epsilon \|u\|^2_{L^2(W)}.$$

This last inequality follows by integrating $\int W|\nabla u|^2$ once by parts and using $|\nabla W| \leq CW$ and Young's inequality. Exponential decay of W allows removal of the cutoffs. The bounded zeroth order coefficient therefore gives

$$\frac{\mathrm{d}}{\mathrm{d}s}\|u(s)\|^2_{L^2(W)} \leq C\|u(s)\|^2_{L^2(W)}.$$

Gronwall's inequality, followed by $\varepsilon \downarrow 0$, proves $u = 0$, because the zero C_0 initial trace implies a zero weighted L^2 trace. This also proves uniqueness for the inhomogeneous construction by subtraction.

Undoing the dilation now gives a strongly continuous homogeneous semigroup $S(s)$ on $C_0(\mathbb{R}^m)$ for (5.4); concatenation is justified by the uniqueness just proved. Dilation is strongly continuous on C_0, and the equation is autonomous. On $0 \leq s \leq 1$ one has $\|S(s)\| \leq C$. Inhomogeneous evolution for a continuous C_0 source is

$$v(\tau_0 + s) = S(s)v(\tau_0) + \int_0^s S(s-r) f^{\#}(\tau_0 + r)\,\mathrm{d}r. \tag{5.12}$$

The formula follows either from the distributional equation and uniqueness or by approximation with smooth sources. Rotation and complex conjugation preserve the equation and the construction. In particular $S(s)$ preserves E and its real subspace.

5.3 Compactness at positive time

Write $P = S(1)|_E$, reserving this symbol here for a bounded operator, and let P_c be the corresponding operator with $A_f = c^2$ and $B_f = 0$. We prove

$$P - P_c : E \longrightarrow E \quad \text{is compact.} \tag{5.13}$$

Both time-one evolutions map their unit balls to locally equicontinuous functions by the interior estimates. It remains to prove uniform decay of their difference at spatial infinity.

Suppose this decay fails. There are $d_j \in E$ with $\|d_j\|_\infty \leq 1$, points ζ_j with $|\zeta_j| \to \infty$, and $\epsilon_0 > 0$ such that $|(P - P_c)d_j(\zeta_j)| \geq \epsilon_0$. In the moving coordinates (5.5), translate by $\eta_j = e^{-1/4}\zeta_j$. The two solutions, denoted by u_j and $u_{c,j}$ after translation, have the same initial value $d_j(\eta + \eta_j)$ and uniform supremum bounds. Their coefficients converge, smoothly on compact subsets of $\mathbb{R}^m \times [0,1]$, to

$$a_c(s) = \frac{c^2 e^{-s}}{16}, \qquad b_c = \tfrac{1}{2}.$$

This follows from (5.3); it also controls time derivatives in the moving coordinates.

After passing to a subsequence, the translated initial functions converge weakly star in $L^\infty(\mathbb{R}^m)$ to some d_∞. Sequential extraction is available because $L^1(\mathbb{R}^m)$ is separable. Interior estimates and a diagonal extraction give smooth local limits $u_\infty, u_{c,\infty}$ for $s > 0$, including convergence of their

values at $s = 1$. Both limits are bounded on the whole slab. To identify their initial traces, for $\psi \in C_c^\infty(\mathbb{R}^m)$ integrate the translated equation:

$$\left| \int u_j(s)\psi - \int d_j(\eta + \eta_j)\psi \right| \le Cs \sup_{j,\, 0\le r\le 1} \left(\|\Delta^2(a_j(\cdot, r)\psi)\|_{L^1} + \|b_j(\cdot, r)\psi\|_{L^1} \right) \le C_\psi s. \tag{5.14}$$

All four spatial derivatives have fallen on the coefficient times the test function. Uniform coefficient bounds make the last constant independent of j. The identity extends down to $s = 0$ using the individual C_0 initial traces. The same estimate holds for $u_{c,j}$. Passing to the limit proves that both limits have distributional initial value d_∞.

Here is the required whole-space uniqueness for these bounded limits; a C_0 initial trace of the limits is not needed. Their difference w solves

$$w_s + a_c(s)\Delta^2 w + \tfrac{1}{2} w = 0$$

and has zero distributional initial value. For a fixed $t_1 \in (0, 1]$ and $\psi \in \mathcal{S}(\mathbb{R}^m)$, use the backward test function

$$\phi(s) = e^{-(t_1 - s)/2} k_{\int_s^{t_1} a_c(r)\, \mathrm{d}r} * \psi, \qquad 0 \le s \le t_1,$$

with $\phi(t_1) = \psi$. It solves $-\phi_s + a_c(s)\Delta^2\phi + \frac{1}{2}\phi = 0$, is continuous in the Schwartz topology, and hence $\int w(s)\phi(s)$ is constant for $s > 0$. One may justify the pairing by compact cutoffs and then use the Schwartz decay. Boundedness of w extends its zero initial trace from compactly supported tests to every Schwartz test; $\phi(s) \to \phi(0)$ in L^1 also permits this changing test. The constant pairing is therefore zero. Since ψ is arbitrary, $w(t_1) = 0$. This contradicts $|u_\infty(0, 1) - u_{c,\infty}(0, 1)| \ge \epsilon_0$.

We have proved uniform vanishing of the tails of $(P - P_c)$ on the unit ball. Together with local equicontinuity, the Arzelà–Ascoli criterion for C_0 proves (5.13): restrict to a large ball, extract there, and use the common small tails to obtain uniform convergence on the whole space. The closed radial subspace is preserved.

5.4 A spectral weight and the two-sided recursion

For the constant coefficient equation, the moving-coordinate formula on an interval of length k is explicit:

$$(P_c^k d)(\zeta) = e^{-k/2}(k_{\theta_k} * d)(e^{-k/4}\zeta), \qquad \theta_k = \frac{c^2}{16}(1 - e^{-k}), \qquad k \ge 1. \tag{5.15}$$

The L^1 norm of k_θ is independent of $\theta > 0$ by scaling. Thus $\|P_c^k\| \le Ce^{-k/2}$, and the Neumann resolvent series converges for $|\lambda| > e^{-1/2}$.

We spell out the spectral consequence of (5.13). On the connected domain $\Omega = \{|\lambda| > e^{-1/2}\}$ factor

$$\lambda I - P = \left[I - (P - P_c)(\lambda I - P_c)^{-1} \right] (\lambda I - P_c).$$

The operator subtracted from the identity is compact and analytic in λ. The analytic Fredholm alternative implies that $(\lambda I - P)^{-1}$ is meromorphic on Ω, with finite rank principal parts, since it is invertible for large $|\lambda|$. One way to obtain this alternative is to approximate the compact operator at any one parameter by a finite rank operator, invert the remaining operator of norm less than one in a neighborhood, and reduce invertibility to a finite-dimensional analytic determinant. The required approximation is available here: identify radial E with $C_0([0, \infty))$, and use bounded polygonal interpolation operators with compact support, which converge strongly to the identity and hence uniformly on compact subsets of E. The determinant either vanishes identically or

has isolated zeros; continuation through overlapping neighborhoods and invertibility at a large parameter exclude the first alternative. The same reduction makes every coefficient of a pole finite rank. It follows that the spectrum in Ω consists of isolated eigenvalues of finite algebraic multiplicity, with no accumulation inside Ω; these are the usual compact-operator and resolvent arguments of analytic Fredholm theory.

Choose

$$e^{-1/2} < q < 1, \qquad \{|\lambda| = q\} \cap \operatorname{spec}(P) = \varnothing, \qquad \gamma = -\log q \in (0, \tfrac{1}{2}), \qquad \delta = 2\gamma. \tag{5.16}$$

Such a circle exists by the discreteness just proved. There are only finitely many eigenvalues outside it: they lie in a bounded closed annulus contained in Ω. Let Π_u be their summed Riesz projection, defined by resolvent integration around these eigenvalues, and put $\Pi_s = I - \Pi_u$. The resolvent identity gives commuting projections and the invariant decomposition

$$E = E_s \oplus E_u, \qquad E_s = \Pi_s E, \quad E_u = \Pi_u E, \qquad P_s = P|_{E_s}, \quad P_u = P|_{E_u}.$$

The space E_u is finite dimensional, P_u is invertible, and the two restricted spectra lie strictly inside and strictly outside the chosen circle, respectively. If $E_u = \{0\}$, all terms involving it below are omitted. Compactness of the restricted spectra and the spectral radius formula give numbers $r_s < q < r_u$ such that

$$\|P_s^k\| \le C r_s^k, \qquad \|P_u^{-k}\| \le C r_u^{-k}, \qquad k \ge 0. \tag{5.17}$$

For example choose r_s strictly between $r(P_s)$ and q, and r_u strictly between q and the smallest modulus of an eigenvalue of P_u; the spectral radius formula controls all sufficiently large powers, and their finitely many predecessors are absorbed in C. The projections commute with complex conjugation because the enclosed spectral sets are invariant under conjugation.

Fix this δ. If $M = \|f\|_Y$, the source weight gives the exact estimate

$$|f^{\#}(\zeta, \tau)| \le M\rho^{\delta}(1 + |\zeta|^2)^{-3+\delta} = Me^{-\gamma\tau}(1 + |\zeta|^2)^{-3+\delta}. \tag{5.18}$$

Indeed $\sigma^2 = \rho(1 + |\zeta|^2)$ and $\rho^3\sigma^{p_0-4} = \rho^{\delta}(1 + |\zeta|^2)^{-3+\delta}$. This proves that $f^{\#}(\tau) \in E$. It also proves continuity with values in E: on any compact τ interval its spatial tails are uniformly small by (5.18), while on a fixed spatial ball local Hölder continuity of f and smoothness of the change of coordinates give uniform continuity in τ.

Let the one-step source increments be

$$l_j = \int_0^1 S(1 - s) f^{\#}(j + s)\, \mathrm{d}s, \qquad j \in \mathbb{Z}.$$

Then $\|l_j\|_E \le CMq^j$. Define

$$v_j = \sum_{k=0}^{\infty} P_s^k \Pi_s l_{j-1-k} - \sum_{k=0}^{\infty} P_u^{-k-1} \Pi_u l_{j+k}. \tag{5.19}$$

Both series converge absolutely for every integer j. More explicitly, their norms are at most

$$CMq^j \left[q^{-1} \sum_{k=0}^{\infty} (r_s/q)^k + r_u^{-1} \sum_{k=0}^{\infty} (q/r_u)^k \right] \le C' M q^j.$$

Shifting the summation indices proves $v_{j+1} = Pv_j + l_j$. On $[j, j+1]$ define v by (5.12) with initial value v_j. The definitions agree at the endpoints and give

$$\|v(\tau)\|_E \le CMe^{-\gamma\tau}, \qquad |V(z,t)| \le CM\rho^{-1+\delta}. \tag{5.20}$$

The continuous joining creates no distributional time-boundary term, so (5.4) holds across every joined face. Interior regularity makes it a classical equation there. The construction is linear and real on real sources. The second sum in (5.19) selects the outside spectral components using future forcing; its use is permitted precisely because no initial values have been imposed.

5.5 The full weighted parabolic estimate

We finish by strengthening (5.20) to the X estimate. This step uses $\delta > 0$ as well as $\delta < 1$. For a box centered at (z_0, t_0) with $r_0^2 = |z_0|^2 \le \rho_0$, one has $\sigma_0^2 \asymp \rho_0$. On a fixed enlargement of this box, chosen small in units (σ_0, σ_0^4), the time stays strictly negative, $\rho \asymp \rho_0$, and $|z|^2/\rho$ stays bounded. In these units the coefficients A and $\sigma_0^4 B$ have bounded derivatives of every fixed order, and the principal coefficient has a uniform positive lower bound. The source has scaled C^α norm at most CM, after division by $\sigma_0^{p_0-4}$. Also

$$|V| \le CM\rho_0^{-1+\delta} \le C'M\sigma_0^{p_0}.$$

Lemma 4.2, with the fixed preceding-time enlargement, therefore gives the required full X bound on every such box.

For the remaining boxes a separate estimate restores the spatial weight. Fix $a > 0$ and write

$$z = ay, \qquad t = a^4 s, \qquad U(y,s) = a^{-p_0} V(ay, a^4 s), \qquad g(y,s) = a^{4-p_0} f(ay, a^4 s).$$

Use the fixed annuli

$$\mathcal{U} = \{\tfrac{1}{2} < |y| < 2\}, \qquad \mathcal{U}' = \{\tfrac{3}{4} < |y| < \tfrac{5}{4}\}.$$

On $\mathcal{U} \times (-4, 0)$ the rescaled equation is

$$U_s + \frac{G(|y|^2/\sqrt{-s})^2}{16}\Delta_y^2 U + \mathfrak{b}(y,s)U = g, \qquad \mathfrak{b}(y,s) = (-s)^{-1} B_f\big(y/(-s)^{1/4}\big). \tag{5.21}$$

All coefficients are independent of a. Their smooth bounds remain uniform as $s \uparrow 0$. To verify the assertion for the zeroth order term explicitly, put $Y = |y|^2/\sqrt{-s}$ and use

$$\mathfrak{b}(y,s) = -(-s)^{-1}\frac{YG'(Y)}{G(Y)}.$$

The expansion of G in powers of Y^{-2} gives an expansion of $-YG'/G$ starting at Y^{-2}. Since $Y^{-2} = (-s)/|y|^4$, division by $-s$ leaves a smooth expansion in nonnegative integral powers of $-s$. Differentiated remainders of arbitrarily high order show that both this coefficient and the principal coefficient extend smoothly to $s = 0$ on the annulus. Away from $s = 0$ their bounds follow from ordinary smoothness of G. The principal coefficient stays uniformly positive throughout.

Furthermore

$$\frac{\sigma(ay, a^4 s)}{a} = \big(|y|^2 + \sqrt{-s}\big)^{1/2}$$

is bounded above and below on the larger annular slab. A fixed covering by the local boxes of Definition 4.1 therefore gives $\|g\|_{C^\alpha(\mathcal{U}\times(-4,0))} \le CM$. Here and below the norm is parabolic; pairs outside a common small box are controlled by the supremum bound and their fixed positive separation.

The preliminary estimate becomes

$$|U(y,s)| \le CM(-s)^{(-1+\delta)/2},$$

with no factor of a: indeed $a^{-p_0}(a^2\sqrt{-s})^{-1+\delta} = (-s)^{(-1+\delta)/2}$. Consequently

$$\|U\|^2_{L^2(\mathcal{U}\times(-4,0))} \le CM^2|\mathcal{U}| \int_0^4 q^{-1+\delta}\,\mathrm{d}q = CM^2|\mathcal{U}|\frac{4^\delta}{\delta} < \infty. \tag{5.22}$$

The value of δ has already been fixed, so the dependence on $1/\delta$ is harmless.

Apply Lemma 4.3 to (5.21) on nested annuli, with the preceding-time buffer from -4 to -2. It gives

$$\sum_{j=0}^{4} \|D_y^j U\|_{C^\alpha(\mathcal{U}'\times(-2,0))} + \|U_s\|_{C^\alpha(\mathcal{U}'\times(-2,0))} \le CM. \tag{5.23}$$

One first applies the estimate with terminal time $b < 0$. On each such truncated slab U has finite supremum norm by (5.20); the coefficient, source, and L^2 bounds above are independent of b. Letting $b \uparrow 0$ then proves (5.23), including Hölder quotients between times arbitrarily close to zero. The estimate uses preceding times and requires no terminal value at $s = 0$.

Finally take an arbitrary remaining center with $r_0^2 > \rho_0$ and choose $a = r_0$. Its rescaled time satisfies $-1 < s_0 < 0$ and $1 < \sigma_0/a < \sqrt{2}$. The entire defining X box lies in $\mathcal{U}' \times (-2, 0)$ when the fixed box radius a_0 is small: its spatial radius is at most $\sqrt{2}a_0$ and its time radius at most $4a_0^4$. In particular a box near the dividing time $s = -1$ has the required preceding-time buffer. Rescaling (5.23) gives its weighted X norm at most CM. The core and annular cases cover all centers. Thus $\|V\|_X \le C\|f\|_Y$, completing the proof of Theorem 5.1.

6 The localized nonlinear construction

Fix $n = 7$ and the profile G from Theorem 3.1. Put $v_* = x(1-x)$, $\Sigma = S^{23}$, $r = \sqrt{x}$, $\rho = \sqrt{-t}$ and $\sigma = (x+\rho)^{1/2}$. All functions are invariant under $O(18) \times O(6)$. Let $0 \le \chi \le 1$ be smooth on $[0,\infty)$, equal to one on $[0,1]$ and zero on $[2,\infty)$. For $0 < \lambda < 1/8$ define

$$\chi_\lambda(x) = \chi(x/\lambda), \qquad F_\lambda = \chi_\lambda F, \qquad b_\lambda = 1 + v_* F_\lambda.$$

Set $\mu = \min(1, \inf G) > 0$ and $M = \max(1, \sup G)$. The identity

$$b_\lambda = 1 - \chi_\lambda + \chi_\lambda\{x + (1-x)G(x/\rho)\} \quad \Longrightarrow \quad \mu \le b_\lambda \le M \tag{6.1}$$

is uniform in all parameters. Define

$$P_\lambda = \partial_t + b_\lambda^2 \mathcal{A} + 2v_* b_\lambda \mathcal{A} F_\lambda, \tag{6.2}$$

$$E_\lambda = (F_\lambda)_t + b_\lambda^2 \mathcal{A} F_\lambda. \tag{6.3}$$

Choose $0 < \alpha < \alpha' < 1$ and the number $\delta \in (0,1)$ from Theorem 5.1. The spaces X, X', Y have weight $p_0 = -2 + 2\delta$ and scale σ as in Definition 4.1.

Proposition 6.1 (Localized correction)**.** *For each fixed $0 < \lambda < 1/8$, each sufficiently small $e > 0$, and each $T_0 > 0$, there exist $0 < T < T_0$ and a real invariant $V \in X_\alpha$ on $\Sigma \times [-T, 0)$ such that*

$$\|V\|_{X'_\alpha} \le e, \qquad H = F_\lambda + V, \tag{6.4}$$

$$H_t + (1 + v_* H)^2 \mathcal{A} H = 0, \qquad \mu/2 \le 1 + v_* H \le M + \mu/2. \tag{6.5}$$

The function H is smooth on $\Sigma \times (-T, 0)$. The upper bound allowed for e may depend on λ; the positivity bounds in (6.5) *do not.*

6.1 Scaled coefficients and the cutoff error

On a box of center scale s, write $O_{\mathrm{sc}}(s^q)$ for a smooth quantity whose derivatives of spatial order j and temporal order k are bounded by $C_{j,k}s^{q-j-4k}$. The scale at the center is held fixed during differentiation. These bounds imply scaled parabolic Hölder estimates at both exponents α, α'.

Lemma 6.2. *If $T \le \lambda^2$, then*

$$F_\lambda = O_{\mathrm{sc}}(\sigma^{-2}), \quad \mathcal{A}F_\lambda = O_{\mathrm{sc}}(\sigma^{-6}), \quad b_\lambda^2 = O_{\mathrm{sc}}(1), \quad 2v_* b_\lambda \mathcal{A}F_\lambda = O_{\mathrm{sc}}(\sigma^{-4}).$$

The constants in these coefficient estimates can be chosen independently of λ, T. The residual satisfies

$$E_\lambda = \begin{cases} O_{\mathrm{sc}}(\sigma^{-4}), & x \le \lambda, \\ O_{\mathrm{sc}}(\sigma^{-6}), & \lambda \le x \le 2\lambda, \\ 0, & x \ge 2\lambda. \end{cases} \tag{6.6}$$

On the inner region, if $P_\lambda - \widehat{P} = \sum d_\nu D^\nu$, then $\sigma^{4-|\nu|} d_\nu = O_{\mathrm{sc}}(\sigma^2)$ in scaled units. For fixed λ, the coefficients extend smoothly to $t = 0$ away from $x = 0$.

Proof. For $x \lesssim \rho$, differentiate $F = \rho^{-1} h(x/\rho)$ using $x = |z|^2$. For $x \gtrsim \rho$, the differentiated expansion of G gives, to arbitrary finite order,

$$F(t,x) = \frac{c-1}{x} + \sum_{j=1}^{N} g_j (-t)^j x^{-2j-1} + \text{differentiated remainder}. \tag{6.7}$$

Both regions give $F = O_{\mathrm{sc}}(\sigma^{-2})$. On the cutoff annulus $\sigma \asymp \sqrt{\lambda}$, because $\rho \le \lambda$, and $D_z^j \chi_\lambda = O(\lambda^{-j/2})$. Thus multiplication by the cutoff preserves these scaled bounds, including all time derivatives.

Near the core, the coefficients in $\mathcal{A} - \widehat{\mathcal{A}}$ of spatial orders four, three and two are respectively $O(r^2), O(r), O(1)$, by Lemma 2.2; lower coefficients are smooth. After rescaling they all acquire the factor $O(\sigma^2)$. Also $v_* - x = -x^2 = O_{\mathrm{sc}}(\sigma^4)$ and $b_\lambda - G = -x^2 F = O_{\mathrm{sc}}(\sigma^2)$ wherever the cutoff is one. For the zeroth-order coefficient, write

$$2v_* b_\lambda \mathcal{A}F - 2xG\widehat{\mathcal{A}}F = 2(v_* b_\lambda - xG)\widehat{\mathcal{A}}F + 2v_* b_\lambda (\mathcal{A} - \widehat{\mathcal{A}})F.$$

Each term is $O_{\mathrm{sc}}(\sigma^{-2})$, which is precisely the claimed mismatch after multiplication by σ^4. The exact flat equation then gives the first line of (6.6). On the annulus use the un-cancelled fourth-order bound, giving the second line. The last line follows from $F_\lambda = 0$. Formula (6.7) proves the assertion at the terminal time. □

In particular $\|E_\lambda\|_{Y_\alpha}$ need not tend to zero with λ: its annular contribution is bounded by a constant times $\lambda^{-\delta}$. The next two lemmas concern a fixed λ. Constants written C_λ are finite before any inner radius or time interval is chosen.

6.2 A right inverse with uniform inner parameters

Lemma 6.3. *For fixed λ, there are $\beta_0, \eta_0, C_D > 0$ such that, whenever $0 < \beta \le \beta_0$ and $0 < T \le \eta_0 \beta^4$, P_λ has a right inverse $D_\lambda : Y_\alpha \to X_\alpha$ with norm at most C_D. These constants are independent of β, T. No initial value is imposed on $D_\lambda f$.*

Proof. Take $\beta_0 < \sqrt{\lambda}/10$. Choose fixed smooth functions of r/β with the following properties:

	equal to one	equal to zero
χ_i	$r \le \beta$	$r \ge 2\beta$
χ_b	$r \le 2\beta$	$r \ge 4\beta$
χ_o	$r \ge \beta$	$r \le \beta/2$
κ	$r \ge \beta/2$	$r \le \beta/4$.

They are smooth at the singular orbit because they are constant there. The support of χ_b lies strictly inside $x < \lambda$.

Extend $f_i = \chi_i f$ to a radial source on $\mathbb{R}^{18} \times (-\infty, 0)$ by zero outside the core chart and, for $t < -T$, by the formula

$$E_i f(z,t) = \zeta\Big(\frac{-T-t}{s_T(z)^4}\Big) f_i(z,-T), \quad t < -T, \qquad s_T(z) = (|z|^2 + \sqrt{T})^{1/2}, \tag{6.8}$$

where $\zeta = 1$ near zero and $\zeta = 0$ on $[1,\infty)$. On its support $\sigma(z,t) \asymp s_T(z)$. The functions s_T have uniformly controlled derivatives in these units, and multiplication by χ_i has the same property on its transition. The trace and cross-face Hölder estimates follow by inserting the value at $t = -T$. This proves the required extension bound. We use (6.8) throughout.

Let $W = \widehat{D} E_i f$ and $V_i = \chi_b W$. Then

$$P_\lambda V_i = f_i + R_i f + J_i f, \tag{6.9}$$

$$R_i f = (P_\lambda - \widehat{P})(\chi_b W), \qquad J_i f = [\widehat{P}, \chi_b] W, \tag{6.10}$$

$$\|R_i f\|_Y \le C\beta^2 \|f\|_Y, \qquad \|J_i f\|_Y \le C\|f\|_Y. \tag{6.11}$$

For the first bound use $\sigma \lesssim \beta$ on the support and $T \le \beta^4$; for the second, every cutoff derivative is balanced by its scaled derivative weight. The commutator is supported in $2\beta \le r \le 4\beta$.

Put $s_\beta = (r^2 + \beta^2)^{1/2}$, an equivalent smooth choice for $\max(r,\beta)$, and replace the coefficients of P_λ by

$$P_{\lambda,\beta} = \partial_t + [\kappa b_\lambda^2 + (1-\kappa)]\mathcal{A} + \kappa(2v_* b_\lambda \mathcal{A} F_\lambda).$$

It equals P_λ where $r \ge \beta/2$. Its principal multiplier is bounded away from zero. All coefficients, with the zeroth-order one multiplied by s_β^4, have uniform scaled bounds at both Hölder exponents. Indeed inside $r \le \beta/4$ they are constant; on the transition this is Lemma 6.2 with $\sigma \asymp \beta$; outside it $\sigma \asymp s_\beta$. At the localization annulus $r \asymp \sqrt{\lambda}$, the same lemma applies. Thus none of these bounds worsens as β, T decrease.

Let $f_o = f - f_i - J_i f$. It vanishes on $r < \beta$. Solve $P_{\lambda,\beta} Z = f_o$, $Z(-T) = 0$, on the entire sphere. Lemma 4.4 and scale comparison on the source support give

$$\|Z\|_{X_\alpha(s_\beta)} \le C_\lambda \|f\|_Y, \qquad \sup s_\beta^{-p_0}|Z| \le C_\lambda \eta \|f\|_Y, \quad \eta = T/\beta^4.$$

For $V_o = \chi_o Z$, the exact identity is

$$P_\lambda V_o = f_o + J_o f, \qquad J_o f = [P_{\lambda,\beta}, \chi_o] Z.$$

The commutator has spatial order at most three and is supported where $\sigma \asymp s_\beta \asymp \beta$. Lemma 4.5 therefore gives $\|J_o f\|_Y \le C_\lambda \eta^{(1-\alpha)/4} \|f\|_Y$. Consequently

$$D_{\mathrm{app}} f = V_i + V_o, \quad P_\lambda D_{\mathrm{app}} = I + \mathcal{R}, \quad \mathcal{R} = R_i + J_o, \tag{6.12}$$

$$\|D_{\mathrm{app}}\|_{Y\to X} \le C_\lambda, \qquad \|\mathcal{R}\|_{Y\to Y} \le C_\lambda(\beta^2 + \eta^{(1-\alpha)/4}). \tag{6.13}$$

The inner commutator has canceled exactly. Fix β_0, then η_0, so the final bound is at most $1/2$ throughout the stated range. The Neumann series gives $D_\lambda = D_{\mathrm{app}}(I + \mathcal{R})^{-1}$ and a bound C_D independent of the subsequently chosen β, T. □

6.3 The large residual has a small corrected value

Lemma 6.4. *For the inverses of Lemma 6.3,*

$$\|D_\lambda E_\lambda\|_{X'_\alpha} \le C_\lambda \left(\beta^{2-2\delta} + \eta^{(\alpha'-\alpha)/4} + \eta^{(1-\alpha)/4}\right). \tag{6.14}$$

Proof. Split $E_\lambda = f_c + f_e$ with $f_c = \chi_b E_\lambda$. The support of f_c lies where the residual has order σ^{-4}. Its Y weight gives $\|f_c\|_Y \le C\beta^{2-2\delta}$, hence the first term of (6.14). The complementary source is zero on $r \le 2\beta$. Thus its inner input, ancient correction, $R_i f_e$ and $J_i f_e$ are all zero. In particular $D_{\mathrm{app}} f_e$ is purely an outer zero-data solution times χ_o.

We claim $\|f_e\|_{Y_{\alpha'}(s_\beta)} \le C_\lambda$. In $4\beta \lesssim r \lesssim \sqrt{\lambda}$, its scaled weighted size is at most $Cr^{2-2\delta}$, by the first line of (6.6). In the χ_b transition the same bound holds with $r \asymp \beta$. In the localization annulus it is at most $C\lambda^{-\delta}$. Outside that annulus it is zero. All corresponding scaled derivatives obey these bounds, which prove the claim including boxes crossing support boundaries.

Apply Lemma 4.4 at exponent α' and Lemma 4.5 to this outer solution. The latter includes temporal Hölder seminorms of its fourth spatial derivatives:

$$\|D_{\mathrm{app}} f_e\|_{X'_\alpha} \le C_\lambda \eta^{(\alpha'-\alpha)/4}, \qquad \|\mathcal{R} f_e\|_Y \le C_\lambda \eta^{(1-\alpha)/4}.$$

The identity $D_\lambda(I + \mathcal{R}) = D_{\mathrm{app}}$ gives $D_\lambda f_e = D_{\mathrm{app}} f_e - D_\lambda \mathcal{R} f_e$. The uniform inverse bound now proves (6.14). □

6.4 Contraction and the order of choices

Expanding the exact equation at F_λ gives

$$P_\lambda V = -E_\lambda - N_\lambda(V), \qquad N_\lambda(V) = v_*^2 V^2 \mathcal{A} F_\lambda + (2v_* b_\lambda V + v_*^2 V^2)\mathcal{A} V.$$

On the X'_α ball of radius $e \le 1$, the scaled product inequality and Lemma 6.2 imply

$$\begin{aligned} \|N_\lambda(V)\|_Y &\le C_N e^2, \\ \|N_\lambda(V) - N_\lambda(W)\|_Y &\le C_N e \|V - W\|_{X'_\alpha}. \end{aligned} \tag{6.15}$$

For example $v_*^2 V^2 \mathcal{A} F_\lambda$ has order $e^2 \sigma^{2p_0-2}$; after multiplication by the Y weight σ^{4-p_0} the extra factor is $\sigma^{p_0+2} = \sigma^{2\delta}$, which is bounded. The other quadratic term has the same order; the cubic term has the extra bounded factor $\sigma^{4\delta}$. The coefficient estimates show that C_N is uniform in β, T. No time derivative of V is required to estimate $N_\lambda(V)$.

For clarity the complete order of choices is as follows. Fix λ; determine $C_D, C_N, \beta_0, \eta_0$ in the lemmas. Choose e below the requested size, below a fixed positivity bound, and so that $C_D C_N e \le 1/4$. Next take $\beta \le \beta_0$ so $C_\lambda \beta^{2-2\delta} < e/6$. Finally decrease $\eta \le \eta_0$ until the two remaining terms of (6.14) sum to less than $e/3$ and $T = \eta\beta^4 < \min(T_0, \lambda^2, 1)$. All constants used in this choice were fixed earlier.

The map

$$V \longmapsto -D_\lambda E_\lambda - D_\lambda N_\lambda(V)$$

is a contraction of the closed X'_α ball of radius e. Its fixed point belongs to X_α because the right side does. As $|v_* V| \le Ce\sigma^{2\delta}$, decreasing the fixed positivity bound for e ensures (6.5). Classical interior parabolic regularity, applied successively after taking spatial difference quotients, makes the solution smooth at every $t < 0$. All statements are on the smooth auxiliary sphere, so this also gives endpoint smoothness in x. This proves Proposition 6.1.

7 Uniform geometric estimates and the main proof

The estimates in this section are for the exact solution $H = F_\lambda + V$ of Proposition 6.1. Constants may depend on the fixed profile and dimension, but do not depend on λ, T, e, unless a subscript states otherwise.

7.1 The curvature integral

Expanding K from (2.3) gives

$$\begin{aligned} K(H) = x^2(1-x)^2H'' + 2x(1-x)((n+1)-(n+3)x)H' \\ + \{n(n+1) - 2(n+1)(n+2)x + (n+2)(n+3)x^2\}H. \end{aligned} \tag{7.1}$$

Its coefficients in radial z coordinates of orders two, one, and zero have sizes $O(r^2), O(r), O(1)$ near the core. Therefore the X' bound entails

$$|K(F_\lambda)| \le \frac{C\mathbf{1}_{\{x\le 2\lambda\}}}{x+\rho}, \qquad |K(V)| \le Ce(x+\rho)^{-1+\delta}. \tag{7.2}$$

These estimates also hold at the other endpoint. To check this directly, put $y = 1-x$ in (7.1). The second and first derivative coefficients vanish to orders two and one, respectively. They are smooth differential operators on invariant functions in the transverse Euclidean coordinates at that orbit.

The pushforward of normalized volume is $nx^{n-1}\mathrm{d}x$, independent of the metric, as established in the geometric reduction. For every fixed $1 \le p < n$,

$$\int_0^{2\lambda} \frac{x^{n-1}}{(x+\rho)^p}\,\mathrm{d}x \le \frac{(2\lambda)^{n-p}}{n-p}, \qquad \int_0^1 x^{n-1}(x+\rho)^{-p+p\delta}\,\mathrm{d}x \le \frac{1}{n-p+p\delta}.$$

By Minkowski's inequality,

$$\sup_{-T<t<0} \|S - n(n+1)\|_{L^p(\nu_t)} \le C_p\lambda^{n/p-1} + C_pe. \tag{7.3}$$

At $n = 7, p = 2$ this implies

$$\sup_{-T<t<0} \int (S-56)^2\,\mathrm{d}\nu_t \le C\lambda^5 + Ce^2. \tag{7.4}$$

7.2 Small initial potential in the fixed complex coordinates

At $t_0 = -T/2$ write $u = u_* + \psi$ and choose its affine part by $\psi(1) = \psi'(1) = 0$. Then

$$\psi'' = -\frac{H}{1+v_*H}, \qquad \psi(x) = \int_x^1 (s-x)\psi''(s)\,\mathrm{d}s.$$

Positivity and the X' bound give

$$\|\psi\|_\infty \le \frac{2}{\mu}\int_0^1 s|H(s)|\,\mathrm{d}s \le C(\lambda+e).$$

Indeed $|F_\lambda| \le C\mathbf{1}_{x\le 2\lambda}/(x+\rho)$, and the correction contributes at most $Ce\int_0^1 s(s+\rho)^{-1+\delta}\mathrm{d}s \le Ce$. The Legendre transform is nonexpansive in the uniform norm: if $|u-u_*| \le a$, then the suprema of $x\ell - u(x)$ and $x\ell - u_*(x)$ differ by at most a for every ℓ. The globally smooth potential furnished by the geometric reduction thus obeys

$$\|\phi_0\|_{C^0} \le C(\lambda+e). \tag{7.5}$$

This normalization is made only at t_0. Afterwards the affine part evolves by $u_t = K(H)$, so the resulting metrics satisfy Calabi flow in the fixed complex coordinates.

7.3 One Sobolev constant for all the examples

In momentum and Hopf coordinates, the real metric, up to fixed common convention factors, is

$$g_v = \frac{\mathrm{d}x^2}{4v} + v\eta^2 + x g_{\mathbb{CP}^{n-1}}, \qquad v = v_* b, \quad b = 1 + v_* H. \tag{7.6}$$

The three summands are orthogonal. Identify the momentum x and the angular coordinates with those of the reference metric. This identification defines a smooth diffeomorphism for each $t < 0$, including the two special orbits. For example, $q = x(1-x)^{-1}e^{\psi'(x)}$ from (2.8) shows that the radial coordinate change multiplies the reference radius by the positive smooth factor $e^{\psi'(x)/2}$; at infinity use the reciprocal radius. The inverse is smooth by the same formula.

After this identification, the ratios of the three metric components are $b^{-1}, b, 1$. Therefore

$$\Lambda^{-1} g_* \le g_v \le \Lambda g_*, \qquad \Lambda = \max\{1, 2/\mu, M + \mu/2\}. \tag{7.7}$$

The volume forms in these coordinates agree exactly: the radial and Hopf factors cancel, leaving a constant multiple of $x^{n-1}\mathrm{d}x$ times the fixed angular measure. Apply the fixed Sobolev inequality of g_* to any smooth function under the identification. Its $L^{2n/(n-1)}$ and L^2 norms are unchanged, and its reference Dirichlet integral is at most Λ times the g_v Dirichlet integral. Thus

$$C_S(g_v) \le \Lambda C_S(g_*). \tag{7.8}$$

The diffeomorphisms need not have uniformly bounded derivatives in the original complex coordinates. Such a bound is not used in the invariant inequality (7.8).

7.4 Completion of the proof

Fix $2 \le p < 7$ and $\varepsilon, \tau > 0$. Choose λ sufficiently small that the first term in (7.3) is less than $\varepsilon/2$ and the λ term in (7.5) is less than $\varepsilon/2$. Choose e to meet the contraction and positivity requirements of Proposition 6.1, and so the remaining terms in those two inequalities are less than $\varepsilon/2$. Apply that proposition with $T_0 = 2\tau$.

At the center the localization cutoff equals one, so

$$H(t,0) = \rho^{-1}G'(0) + O(e\rho^{-1+\delta}), \qquad S(t,o) = -56G'(0)\rho^{-1} + O(1 + e\rho^{-1+\delta}).$$

Let $t_0 = -T/2$ and translate time by $T/2$. The exact geometric reduction gives a smooth Calabi flow on $[0, T/2)$. Smooth uniqueness, whose energy argument was given in Lemma 2.3, identifies it with the maximal flow from its initial slice. Its positive scalar-curvature divergence prevents any smooth extension through $T/2$. Hence $T_* = T/2 < \tau$. Equations (7.3), (7.5) and (7.8) establish all the remaining assertions of Theorem 1.1, with $a = -56G'(0) > 0$.

8 The critical exponent

The construction also locates precisely the loss of strong critical integrability. Fix one constructed flow and its cutoff radius. In this section $t < 0$ denotes time before its singularity and $\rho = \sqrt{-t}$. Define

$$B_G(y) = -yG''(y) - 2nG'(y) - n(n-1)(G(y)-1)/y, \qquad A_G = n(n-1)(1-c) > 0.$$

Then $B_G(0) = -n(n+1)G'(0) > 0$ and $B_G(y) = A_G/y + O(y^{-3})$. On $x \le \lambda$ the exact curvature is

$$S - n(n+1) = \rho^{-1}B_G(x/\rho) + O(1) + O(e(x+\rho)^{-1+\delta}). \tag{8.1}$$

The $O(1)$ term comes from the difference between v_*^2 and x^2 in K. On $x \geq \lambda$, the curvature is bounded for this fixed λ by (7.2) and the smooth outer expansion.

Proposition 8.1. *For the constructed seven-dimensional flow, with $R_S = S - 56$,*

$$\int |R_S|^7 \, \mathrm{d}\nu_t = 7A_G^7 \log(1/\rho) + O(1), \qquad \sup_{t<0} \|R_S\|_{L^{7,\infty}(\nu_t)} < \infty. \tag{8.2}$$

For $p > 7$,

$$\int |R_S|^p \, \mathrm{d}\nu_t \sim 7\rho^{7-p} \int_0^\infty |B_G(y)|^p y^6 \, \mathrm{d}y. \tag{8.3}$$

The integral on the right is positive and finite.

Proof. For the model critical integral put $x = \rho y$. Since $|B_G(y)|^7 y^6 = A_G^7/y + O(y^{-3})$ for large y, its integral up to λ/ρ is $A_G^7 \log(1/\rho) + O(1)$. For the perturbation in (8.1), use $||u+v|^7 - |u|^7| \leq C(|u|^6|v| + |v|^7)$. The mixed term involving the correction is bounded by a constant times $\int_0^\lambda x^{-1+\delta} \mathrm{d}x$; the correction's seventh power is also integrable. The bounded geometric error gives an integrable mixed term. On $x \lesssim \rho$ the corresponding terms are bounded by $O(\rho^\delta) + O(\rho^{7\delta}) + O(\rho)$. This proves the first assertion of (8.2).

Globally $|R_S| \leq C/x$, hence $\nu_t\{|R_S| > a\} \leq \min(1, (C/a)^7)$. This proves the weak norm bound. It is a fixed finite bound, not an arbitrarily small one. For $p > 7$, $x = \rho y$ and dominated convergence give the model asymptotic. The correction's rescaled p-power integral tends to zero: the three cases $p(1-\delta) > 7$, $= 7$, and < 7 give respectively $O(\rho^{p\delta})$, $O(\rho^{p-7}\log(1/\rho))$, and $O(\rho^{p-7})$. The bounded error is smaller as well. Minkowski's inequality now proves (8.3). □

9 Conclusion

The construction yields smooth Calabi flows in a fixed Kähler class on $\mathbb{CP}^7$ with arbitrarily small initial potentials, arbitrarily small Calabi energy throughout their smooth existence intervals, and one common Sobolev bound, yet with finite-time scalar-curvature blowup. More generally, any prescribed subcritical L^p norm, $2 \leq p < 7$, can be made uniformly small. The essential step is the localized nonlinear correction: its annular residual is handled after applying the outer zero-data inverse, with the cutoff radius fixed before the inner radius and the time interval are chosen. The strong L^7 norm diverges logarithmically, so this example does not decide continuation under a uniform strong L^7 bound. The finite rational certificate verifies the arithmetic assertions used to construct the profile; the analytic and geometric arguments above are the remaining parts of the proof.